

ENGINEERING CHEMISTRY LAB

Engineering Chemistry Lab is a laboratory course that is designed to provide students with practical experience in the application of chemical principles to engineering problems. The course typically includes experiments that demonstrate the principles of chemical thermodynamics, kinetics, electrochemistry, and materials science.

Some of the key objectives of the Engineering Chemistry Lab course are:

1. To provide students with hands-on experience in conducting experiments and analyzing data.
2. To develop students' skills in experimental design, data analysis, and scientific communication.
3. To reinforce the principles of chemical thermodynamics, kinetics, electrochemistry, and materials science through practical applications.
4. To introduce students to modern instrumentation and techniques used in chemical analysis.

The experiments conducted in the Engineering Chemistry Lab course may vary depending on the specific curriculum and the focus of the course. Some common experiments that may be included in the course are:

1. Determination of the heat of reaction using calorimetry.
2. Measurement of reaction rates and the study of reaction kinetics.
3. Investigation of electrochemical cells and the principles of electrochemistry.
4. Synthesis and characterization of materials, such as polymers and nanomaterials.

The Engineering Chemistry Lab course is an important component of many engineering programs, as it provides students with the opportunity to apply the principles of chemistry to real-world problems and to develop the skills necessary for success in their future careers.

Course Objectives:

1. To provide a comprehensive understanding of the subject matter.
2. To develop critical thinking and analytical skills.
3. To enhance the ability to apply theoretical concepts to practical situations.
4. To encourage independent learning and self-motivation.
5. To foster effective communication and collaboration skills.
6. To promote ethical and responsible behavior in the field of study.
7. To prepare students for further academic or professional pursuits in the subject area.

1. To enable the students to understand about the fundamental concepts of analytical instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are commonly used in scientific research, quality control, and medical diagnosis.

Some fundamental concepts of analytical instruments include:

- **Accuracy:** The degree to which the measurement result agrees with the true value of the quantity being measured.
- **Precision:** The degree to which repeated measurements of the same quantity agree with each other.
- **Sensitivity:** The smallest change in the quantity being measured that can be detected by the instrument.
- **Resolution:** The smallest difference between two measurements that can be distinguished by the instrument.
- **Calibration:** The process of adjusting the instrument to ensure that its measurements are accurate and precise.
- **Signal-to-noise ratio:** The ratio of the strength of the signal being measured to the strength of the background noise.

Understanding these concepts is essential for students to effectively use and interpret the results of analytical instruments. It is also important for students to learn about the different types of analytical instruments and their applications in various fields.

Analysis of Chloride Content, Hardness, and Alkalinity of Water

1. **Chloride Content:** Chloride content in water is an important parameter to measure as it can indicate the presence of contamination from human or animal waste, industrial pollution, or saltwater intrusion. The chloride content of water can be measured using titration methods or by using a chloride ion-selective electrode.
2. **Hardness:** Hardness in water is caused by the presence of dissolved minerals, primarily calcium and magnesium. Hard water can cause scaling in pipes and appliances, and can also interfere with the effectiveness of soaps and detergents. The hardness of water can be measured using titration methods or by using a water hardness test kit.
3. **Alkalinity:** Alkalinity in water refers to its ability to neutralize acids. It is an important parameter to measure as it can affect the pH of the water and its ability to support aquatic life. Alkalinity can be measured using titration methods or by using a pH meter.

In summary, the analysis of chloride content, hardness, and alkalinity of water

is important for understanding the quality of water and its potential impacts on human health and the environment. These parameters can be measured using a variety of methods, including titration, ion-selective electrodes, test kits, and pH meters.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- **pH:** pH is a measure of the acidity or basicity of a liquid. It is defined as the negative logarithm of the hydrogen ion concentration. The pH scale ranges from 0 to 14, with 7 being neutral. A pH less than 7 is acidic, while a pH greater than 7 is basic. pH can be measured using a pH meter or pH paper.
- **Surface tension:** Surface tension is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the liquid molecules. Surface tension can be measured using a number of methods, including the drop weight method, the maximum bubble pressure method, and the Wilhelmy plate method.
- **Viscosity:** Viscosity is a measure of a liquid's resistance to flow. It is caused by the internal friction between the molecules of the liquid. Viscosity can be measured using a number of methods, including the capillary tube method, the falling ball method, and the rotational viscometer method.

These properties of a liquid are important to understand as they can affect the behavior of the liquid in various applications. For example, the pH of a liquid can affect its reactivity, the surface tension can affect its ability to wet a surface, and the viscosity can affect its ability to flow. Understanding these properties can help in the selection and use of liquids in various applications.

4. To enable the students to understand about the preparation of different resins.

Resins are solid or highly viscous substances that are typically convertible into polymers. They are usually mixtures of organic compounds, mainly terpenes and their derivatives. Here are some points to consider when preparing different resins:

1. **Types of resins:** There are several types of resins, including natural resins (such as amber, copal, and dammar) and synthetic resins (such as epoxy, polyurethane, and silicone). Each type of resin has its own unique properties and preparation methods.

2. **Raw materials:** The raw materials used to prepare resins vary depending on the type of resin being prepared. For example, natural resins are typically derived from plant or animal sources, while synthetic resins are typically derived from petrochemicals.
3. **Preparation methods:** The methods used to prepare resins vary depending on the type of resin being prepared. For example, natural resins are typically extracted from their sources using solvents, while synthetic resins are typically prepared through chemical reactions.
4. **Safety precautions:** When preparing resins, it is important to follow proper safety precautions to avoid injury or harm. This may include wearing protective clothing and equipment, working in a well-ventilated area, and following proper handling and disposal procedures.
5. **Quality control:** To ensure that the prepared resins meet the desired specifications, it is important to implement proper quality control measures. This may include testing the resins for properties such as viscosity, hardness, and chemical resistance.

In summary, the preparation of different resins involves understanding the types of resins, the raw materials used, the preparation methods, the safety precautions, and the quality control measures. By following these guidelines, students can learn how to prepare different resins effectively and safely.

5. Synthesis of Organic Compounds: Adipic Acid and Paracetamol

Adipic acid and paracetamol are two important organic compounds that can be synthesized through both conventional and green routes. Here, we will discuss the synthesis of these compounds through both methods.

Adipic Acid

Adipic acid is an important dicarboxylic acid used in the production of nylon. It can be synthesized through both conventional and green routes.

Conventional Route

The conventional route for the synthesis of adipic acid involves the oxidation of cyclohexane with nitric acid. This process produces nitrous oxide, a potent greenhouse gas, as a byproduct.

Green Route

The green route for the synthesis of adipic acid involves the oxidation of cyclohexene with hydrogen peroxide in the presence of a tungsten catalyst. This

process produces water as the only byproduct, making it a more environmentally friendly option.

Paracetamol

Paracetamol is a widely used analgesic and antipyretic drug. It can also be synthesized through both conventional and green routes.

Conventional Route

The conventional route for the synthesis of paracetamol involves the nitration of phenol followed by the reduction of the nitro group to an amino group and the acetylation of the amino group.

Green Route

The green route for the synthesis of paracetamol involves the direct acetylation of para-aminophenol with acetic anhydride in the presence of a solid acid catalyst. This process produces fewer byproducts and is considered a more environmentally friendly option.

In conclusion, the synthesis of organic compounds such as adipic acid and paracetamol can be achieved through both conventional and green routes. The green routes offer more environmentally friendly options with fewer byproducts. It is important for students to understand these methods and their implications for the environment.

LIST OF EXPERIMENTS

1. **The Milgram Experiment:** A series of social psychology experiments conducted by Yale University psychologist Stanley Milgram in the 1960s, which measured the willingness of study participants to obey an authority figure who instructed them to perform acts that conflicted with their personal conscience.
2. **The Stanford Prison Experiment:** A social psychology experiment conducted at Stanford University in 1971 by Philip Zimbardo, which aimed to investigate the psychological effects of perceived power, focusing on the struggle between prisoners and prison officers.
3. **The Asch Conformity Experiment:** A series of studies conducted in the 1950s by Solomon Asch, which aimed to investigate the extent to which social pressure from a majority group could affect a person to conform.
4. **The Pavlov's Dog Experiment:** A series of experiments conducted by Russian physiologist Ivan Pavlov in the 1890s, which aimed to investigate classical conditioning, a type of learning in which an organism comes to associate stimuli.

5. **The Marshmallow Test:** A study on delayed gratification conducted by Walter Mischel in the 1960s and 1970s, which aimed to investigate the ability of children to delay gratification and the relationship between delayed gratification and future success.
6. **The Hawthorne Effect:** A series of studies conducted in the 1920s and 1930s at the Hawthorne Works factory in Cicero, Illinois, which aimed to investigate the effects of changes in working conditions on worker productivity.
7. **The Bobo Doll Experiment:** A series of studies conducted in the 1960s by Albert Bandura, which aimed to investigate the effects of observing aggressive behavior on the behavior of children.
8. **The Bystander Effect:** A social psychology phenomenon in which individuals are less likely to offer help to a victim when other people are present, first demonstrated in a series of experiments conducted by John Darley and Bibb Latané in the 1960s.
9. **The Little Albert Experiment:** A study conducted by John B. Watson and Rosalie Rayner in 1920, which aimed to investigate classical conditioning in humans by demonstrating that emotional responses could be conditioned in humans.
10. **The Rorschach Inkblot Test:** A projective psychological test developed by Hermann Rorschach in 1921, which aimed to investigate personality characteristics and emotional functioning by analyzing a person's interpretation of inkblots.

1. Calibration of Analytical Equipment and Apparatus

Calibration is the process of checking and adjusting the accuracy of a measuring instrument by comparing it with a reference standard. It is an essential step in ensuring the accuracy and reliability of analytical equipment and apparatus.

Here are some key points to consider when calibrating analytical equipment and apparatus:

- Calibration should be performed regularly to ensure the accuracy of the equipment.
- The frequency of calibration depends on the usage and the stability of the equipment.
- A calibration schedule should be established and followed to ensure that the equipment is calibrated at appropriate intervals.
- Calibration should be performed by trained personnel using appropriate reference standards.
- The results of the calibration should be documented and records should be maintained.

- Any deviations from the expected results should be investigated and corrective actions should be taken.
- The equipment should be recalibrated after any maintenance or repair work.

In summary, calibration is an important process that ensures the accuracy and reliability of analytical equipment and apparatus. It should be performed regularly by trained personnel using appropriate reference standards, and the results should be documented and records maintained. Any deviations from the expected results should be investigated and corrective actions taken.

2. Determination of Hardness of water sample by EDTA method

- Hardness of water is determined by the presence of calcium and magnesium ions in water.
- Ethylenediaminetetraacetic acid (EDTA) is a chelating agent that can be used to determine the hardness of water.
- In this method, a standard solution of EDTA is used to titrate a water sample.
- The EDTA binds with the calcium and magnesium ions present in the water, forming a complex.
- An indicator such as Eriochrome Black T is used to signal the end point of the titration.
- The amount of EDTA used in the titration is proportional to the hardness of the water sample.
- The hardness of the water sample can be calculated using the volume of EDTA used and the concentration of the EDTA solution.
- This method is simple, accurate, and widely used for the determination of water hardness.

3. Determination of Alkalinity of water sample

Alkalinity is a measure of the ability of water to neutralize acids. It is an important parameter in determining the quality of water for various uses. Here are the steps to determine the alkalinity of a water sample:

1. Collect a water sample in a clean container.
2. Measure the pH of the water sample using a pH meter or pH test strips.
3. If the pH is above 7, the water is alkaline.
4. To determine the exact alkalinity, titrate the water sample with a standard acid solution, such as sulfuric acid or hydrochloric acid.
5. The volume of acid required to bring the pH of the water sample to a specific value, usually 4.5, is used to calculate the alkalinity of the water sample.
6. The alkalinity is usually expressed in milligrams per liter (mg/L) of calcium carbonate (CaCO_3).

It is important to note that the alkalinity of water can vary depending on the

source and the presence of certain minerals and substances in the water. Regular testing and monitoring of water alkalinity is important to ensure its suitability for various uses.

4. Determination of pH by titrimetric method

Titrimetric method is a common laboratory technique used to determine the pH of a solution. The process involves the following steps:

1. A solution of known concentration, called the titrant, is prepared. This is usually a strong acid or base.
2. The solution to be tested, called the analyte, is measured and placed in a container.
3. A few drops of an indicator are added to the analyte. The indicator is a substance that changes color depending on the pH of the solution.
4. The titrant is slowly added to the analyte while stirring. The indicator changes color when the pH of the solution reaches a certain value.
5. The volume of titrant added is recorded when the color change occurs. This is called the endpoint.
6. The pH of the solution can be calculated using the volume of titrant added and the concentration of the titrant.

This method is accurate and reliable, but it requires careful preparation and execution. It is commonly used in laboratories and in industrial processes where precise pH measurements are required.

5. Determination of surface tension of given liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the liquid molecules. It is defined as the force acting per unit length on the surface of the liquid. There are several methods to determine the surface tension of a given liquid, including:

1. **Capillary rise method:** This method involves measuring the height to which a liquid rises in a capillary tube due to the surface tension of the liquid. The surface tension is then calculated using the formula:
$$\text{Surface tension} = (\text{weight of liquid raised in the capillary} * \text{gravitational acceleration}) / (2 * \pi * \text{radius of capillary} * \text{height of liquid column}).$$
2. **Maximum bubble pressure method:** This method involves measuring the maximum pressure required to blow a bubble from the end of a submerged tube. The surface tension is then calculated using the formula:
$$\text{Surface tension} = (4 * \text{maximum pressure} * \text{radius of tube}) / (3 * \text{radius of bubble}).$$
3. **Drop weight method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is then calculated using the formula: **Surface tension**

= (weight of droplet * gravitational acceleration) / (pi * diameter of tube).

4. **Wilhelmy plate method:** This method involves immersing a thin flat plate into a liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: Surface tension = (force required to detach plate) / (perimeter of plate).
5. **Du Nouy ring method:** This method involves immersing a thin ring into a liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: Surface tension = (force required to detach ring) / (circumference of ring).

These are some of the common methods used to determine the surface tension of a given liquid. Each method has its own advantages and limitations, and the choice of method depends on the specific requirements of the experiment.

6. Determination of Viscosity of a given liquid by viscometer

Viscosity is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate. A viscometer is an instrument used to measure the viscosity of a fluid. There are several types of viscometers, including capillary, rotational, and falling ball viscometers.

To determine the viscosity of a given liquid using a viscometer, the following steps can be followed:

1. Prepare the viscometer by cleaning it and ensuring that it is properly calibrated.
2. Fill the viscometer with the liquid to be tested.
3. Start the viscometer and record the time it takes for the liquid to flow through the viscometer.
4. Use the recorded time and the calibration data for the viscometer to calculate the viscosity of the liquid.

It is important to note that the viscosity of a liquid can be affected by factors such as temperature and pressure. Therefore, it is important to conduct the viscosity measurement under controlled conditions to ensure accurate results. Additionally, it may be necessary to repeat the measurement several times to obtain an average value for the viscosity of the liquid.

7. Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. Its chemical formula is $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$.

3. It is commonly used as a standard solution in redox titrations due to its stability and the ease of preparation.
4. The strength of a ferrous ammonium sulfate solution can be determined by titrating it against a standard solution of potassium permanganate (KMnO_4) using an external indicator.
5. The external indicator used in this titration is potassium ferricyanide ($\text{K}_3\text{Fe}(\text{CN})_6$).
6. The reaction between ferrous ammonium sulfate and potassium permanganate is a redox reaction, where the iron in ferrous ammonium sulfate is oxidized from Fe^{2+} to Fe^{3+} and the manganese in potassium permanganate is reduced from Mn^{7+} to Mn^{2+} .
7. The end point of the titration is indicated by the appearance of a blue color due to the formation of Prussian blue, which is a complex formed between Fe^{3+} and potassium ferricyanide.
8. The strength of the ferrous ammonium sulfate solution can be calculated using the volume of potassium permanganate solution used and the molar concentration of the potassium permanganate solution.

8. Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is an oxidizing agent that can be used to determine the strength of a solution.
2. An internal indicator is used to determine the endpoint of the titration.
3. The internal indicator used in this experiment is diphenylamine sulfonic acid.
4. The diphenylamine sulfonic acid reacts with the excess potassium dichromate to produce a blue color, indicating the endpoint of the titration.
5. The strength of the potassium dichromate solution can be calculated using the volume of the solution used and the concentration of the analyte.
6. This method is commonly used in analytical chemistry to determine the strength of oxidizing agents.

9. Determination of available chlorine in bleaching powder

Bleaching powder, also known as calcium hypochlorite, is a chemical compound with the formula $\text{Ca}(\text{ClO})_2$. It is widely used as a disinfectant and bleaching agent. The available chlorine in bleaching powder refers to the amount of chlorine that can be released as a disinfectant when the powder is dissolved in water.

Here are the steps to determine the available chlorine in bleaching powder:

1. Weigh accurately about 1g of the sample of bleaching powder and transfer it to a conical flask.
2. Add about 20 ml of distilled water to the flask and shake well to dissolve the bleaching powder.

3. Add 1-2 ml of potassium iodide solution (10%) and 2-3 ml of acetic acid (20%) to the flask.
4. Titrate the liberated iodine with sodium thiosulphate solution (0.1 N) using starch as an indicator.
5. The end point is reached when the blue color of the starch-iodine complex disappears.
6. Repeat the titration with a blank solution (without the bleaching powder) to determine the amount of sodium thiosulphate required for the blank.
7. Subtract the volume of sodium thiosulphate required for the blank from the volume required for the sample to obtain the volume of sodium thiosulphate required for the available chlorine in the sample.
8. Calculate the percentage of available chlorine in the sample using the formula: % available chlorine = (Volume of sodium thiosulphate required for sample x Normality of sodium thiosulphate x 35.5) / Weight of sample.

This is the standard procedure for determining the available chlorine in bleaching powder. It is important to follow the steps carefully and accurately to obtain reliable results.

10. Determination of chloride content in water sample

1. Chloride is one of the major anions found in water and wastewater.
2. The chloride content of water can be determined by titration with silver nitrate solution, using potassium chromate as an indicator.
3. The reaction between silver nitrate and chloride ions produces a white precipitate of silver chloride.
4. The end point of the titration is indicated by the formation of a red-brown color, due to the reaction between excess silver ions and chromate ions.
5. The concentration of chloride ions in the water sample can be calculated from the volume of silver nitrate solution used in the titration.
6. This method is known as the Mohr method and is commonly used for the determination of chloride content in water samples.
7. The accuracy of the method can be affected by the presence of other ions, such as bromide and iodide, which also react with silver nitrate.
8. To improve the accuracy of the method, the sample can be treated with nitric acid to remove interfering ions before titration.
9. The chloride content of water is an important parameter in the monitoring of water quality, as high levels of chloride can indicate pollution from sources such as sewage or industrial waste.
10. The maximum allowable concentration of chloride in drinking water is 250 mg/L, according to the World Health Organization.

11. Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol with formaldehyde. Here are the steps involved in the preparation of PF resin:

1. **Phenol and formaldehyde are mixed:** The first step in the preparation of PF resin is to mix phenol and formaldehyde in the presence of an acidic or basic catalyst. The ratio of phenol to formaldehyde is typically between 1:1 and 1:2.
2. **The reaction is heated:** The mixture is then heated to a temperature of around 80-100°C to initiate the reaction. The reaction is exothermic, meaning it releases heat, so the temperature must be carefully controlled to prevent the reaction from getting too hot.
3. **The reaction is allowed to proceed:** The reaction is allowed to proceed for several hours, during which time the phenol and formaldehyde react to form a complex mixture of compounds. The exact composition of the mixture depends on the reaction conditions, such as the temperature, the catalyst used, and the ratio of phenol to formaldehyde.
4. **The resin is isolated:** Once the reaction is complete, the resin is isolated from the reaction mixture. This can be done by pouring the mixture into water, which causes the resin to separate out as a solid. The solid resin can then be filtered off and washed to remove any remaining impurities.
5. **The resin is dried:** The final step in the preparation of PF resin is to dry the resin to remove any remaining water. This can be done by heating the resin in an oven at a temperature of around 100-120°C until it is completely dry.

PF resin is a versatile material that can be used in a wide range of applications, including as an adhesive, a coating, and a molding compound. Its properties, such as its high strength and heat resistance, make it well-suited for use in demanding environments.

12. Preparation of Urea formaldehyde (UF) resin

Urea formaldehyde (UF) resin is a type of thermosetting polymer that is commonly used in the production of adhesives, finishes, and molded objects. Here are the steps involved in the preparation of UF resin:

1. **Reacting urea and formaldehyde:** The first step in the preparation of UF resin is the reaction of urea and formaldehyde in an aqueous solution. The molar ratio of urea to formaldehyde is typically between 1:1.5 and 1:2.5. The reaction is carried out at a pH of 7-8 and a temperature of 70-90°C.

2. **Condensation:** The next step is the condensation of the reaction mixture. This is done by heating the mixture to 90-100°C and adjusting the pH to 4.5-5.5 using an acid catalyst. The condensation reaction results in the formation of low molecular weight pre-polymers.
3. **Molding:** The pre-polymers are then molded into the desired shape and cured at a temperature of 130-160°C. The curing process involves the cross-linking of the pre-polymers to form a three-dimensional network.
4. **Post-curing:** The final step is post-curing, which involves heating the cured resin at a temperature of 80-100°C for several hours. This step helps to improve the properties of the resin, such as its strength and water resistance.

These are the basic steps involved in the preparation of UF resin. The exact conditions and parameters may vary depending on the specific application and desired properties of the final product.

13. Preparation of Adipic acid / Paracetamol

Adipic acid and Paracetamol are two different compounds with different preparation methods.

Adipic acid:

Adipic acid is an organic compound with the formula $(CH_2)_4(COOH)_2$. It is a white crystalline powder and is mainly used in the production of nylon.

1. Adipic acid can be prepared by the oxidation of cyclohexanol or cyclohexanone with nitric acid.
2. Another method for the preparation of adipic acid is the carbonylation of butadiene to produce adiponitrile, which is then hydrolyzed to adipic acid.

Paracetamol:

Paracetamol, also known as acetaminophen, is an over-the-counter pain reliever and fever reducer.

1. Paracetamol can be prepared by the nitration of phenol to produce para-nitrophenol, which is then reduced to para-aminophenol.
2. The para-aminophenol is then acetylated with acetic anhydride to produce paracetamol.

14. Determination of Cell Conductance of a solution

1. Cell conductance is a measure of the ability of a solution to conduct electricity.

2. It is determined by measuring the electrical resistance of the solution between two electrodes.
3. The conductance is the reciprocal of the resistance, and is measured in units of Siemens (S).
4. The cell constant is a factor that takes into account the geometry of the cell and the distance between the electrodes.
5. The cell conductance is calculated by multiplying the measured conductance by the cell constant.
6. The cell conductance is affected by the concentration and mobility of the ions in the solution, as well as the temperature of the solution.
7. The cell conductance can be used to determine the concentration of ions in a solution, or to monitor changes in the solution during a chemical reaction.

15. Determination of Rate constant of hydrolysis of esters

1. The rate constant of hydrolysis of esters can be determined by measuring the concentration of the reactants or products over time.
2. The reaction can be represented by the equation: ester + water → carboxylic acid + alcohol
3. The rate of the reaction can be expressed as: rate = k[ester]
4. The rate constant, k, can be determined by measuring the initial rate of the reaction and the initial concentrations of the reactants.
5. The initial rate can be determined by measuring the change in concentration of one of the products over a short period of time at the beginning of the reaction.
6. The initial concentrations of the reactants can be determined by measuring the concentrations of the reactants before the reaction starts.
7. The rate constant can then be calculated by substituting the values of the initial rate and initial concentrations into the rate equation.
8. The rate constant can also be determined by measuring the change in concentration of one of the reactants or products over time and plotting a graph of concentration versus time.
9. The slope of the graph can be used to calculate the rate constant.
10. The rate constant can also be determined by measuring the time taken for the reaction to reach a certain point, such as half completion, and using this information to calculate the rate constant.

Element detection and identification of functional groups in organic compounds

1. **Element detection:** The detection of elements in organic compounds involves the determination of the presence of certain elements such as carbon, hydrogen, nitrogen, sulfur, and halogens.
2. **Carbon and hydrogen:** Carbon and hydrogen are usually detected by

heating the compound with copper oxide. The carbon in the compound is oxidized to carbon dioxide and the hydrogen is oxidized to water. The carbon dioxide is then absorbed by a solution of calcium hydroxide, forming calcium carbonate, while the water is absorbed by anhydrous copper sulfate, turning it blue.

3. **Nitrogen:** Nitrogen can be detected by the Lassaigne's test, which involves heating the compound with sodium metal. The nitrogen in the compound reacts with the sodium to form sodium cyanide, which can be detected by its reaction with iron(II) sulfate and sodium hydroxide, forming Prussian blue.
4. **Sulfur:** Sulfur can be detected by heating the compound with sodium, forming sodium sulfide. The sodium sulfide can then be detected by its reaction with lead acetate, forming a black precipitate of lead sulfide.
5. **Halogens:** Halogens can be detected by the Beilstein test, which involves heating the compound with copper wire. The halogens in the compound react with the copper to form copper halides, which can be detected by their characteristic flame colors.
6. **Functional groups:** The identification of functional groups in organic compounds involves the determination of the presence of certain characteristic groups of atoms that determine the chemical behavior of the compound. Common functional groups include alcohols, aldehydes, ketones, carboxylic acids, esters, amines, and amides.
7. **Alcohols:** Alcohols can be identified by their reaction with Lucas reagent, forming a cloudy solution, or by their reaction with sodium, forming hydrogen gas.
8. **Aldehydes:** Aldehydes can be identified by their reaction with Tollens' reagent, forming a silver mirror, or by their reaction with Fehling's solution, forming a red precipitate.
9. **Ketones:** Ketones can be identified by their reaction with 2,4-dinitrophenylhydrazine, forming a yellow or orange precipitate.
10. **Carboxylic acids:** Carboxylic acids can be identified by their reaction with sodium bicarbonate, forming carbon dioxide gas, or by their reaction with litmus, turning it red.
11. **Esters:** Esters can be identified by their characteristic fruity smell, or by their reaction with sodium hydroxide, forming a carboxylic acid and an alcohol.
12. **Amines:** Amines can be identified by their reaction with nitrous acid, forming nitrogen gas, or by their reaction with litmus, turning it blue.
13. **Amides:** Amides can be identified by their reaction with nitrous acid, forming nitrogen gas and a carboxylic acid, or by their reaction with litmus, having no effect.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose any 10 experiments from the list provided above.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read and understand the experiments chosen by the instructor.
- Students should prepare for the experiments by reviewing the relevant material and asking questions if necessary.
- By completing the chosen experiments, students will gain valuable hands-on experience and a deeper understanding of the subject matter.

Course Outcomes:

- Define and explain the key concepts and principles of the course subject.
- Demonstrate an understanding of the course material through application and analysis.
- Develop critical thinking and problem-solving skills related to the course subject.
- Communicate effectively in written and oral forms about the course subject.
- Collaborate effectively with others to achieve common goals related to the course subject.
- Demonstrate ethical and professional behavior in the context of the course subject.
- Apply the knowledge and skills gained in the course to real-world situations.
- Evaluate and reflect on personal learning and growth throughout the course.

Upon completion of the course the student should be able to:

1. Demonstrate a comprehensive understanding of the subject matter.
2. Apply the knowledge and skills acquired in the course to solve problems and make informed decisions.
3. Communicate effectively using the terminology and concepts of the subject.
4. Work collaboratively with others to achieve common goals.
5. Critically evaluate information and arguments, and form well-reasoned opinions.

6. Demonstrate ethical and responsible behavior in academic and professional settings.
7. Engage in lifelong learning and professional development.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course. These outcomes are typically aligned with the course objectives and are used to assess the effectiveness of the course in achieving its goals.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is often used to design course outcomes and assessments. The taxonomy consists of six levels, arranged in a hierarchy from lower-order thinking skills to higher-order thinking skills:

1. **Remembering:** The ability to recall or retrieve previously learned information.
2. **Understanding:** The ability to comprehend the meaning of material.
3. **Applying:** The ability to use learned material in new and concrete situations.
4. **Analyzing:** The ability to break down material into its component parts so that its organizational structure may be understood.
5. **Evaluating:** The ability to make judgments about the value of ideas or materials.
6. **Creating:** The ability to put parts together to form a new whole.

By using Bloom's Taxonomy to design course outcomes, instructors can ensure that their courses are challenging and engaging for students, and that they are effectively assessing students' mastery of the material. Additionally, by aligning course outcomes with Bloom's Taxonomy, instructors can help students develop higher-order thinking skills that will serve them well in their future academic and professional pursuits.

Level

- Level is a term that can refer to a relative position or rank within a graded hierarchy.
- In the context of measurements, level can refer to a specific value on a scale or a reference point used to determine the relative height or depth of an object.
- In construction and engineering, a level is a tool used to determine if a surface is horizontal or vertical.
- In gaming, level can refer to a stage or section of a game, or the player's progress or experience within the game.

- In education, level can refer to the degree of difficulty or advancement of a course or program.
- In music, level can refer to the volume or intensity of sound.
- In computing, level can refer to the degree of access or privilege granted to a user or process within a system.
- In linguistics, level can refer to the degree of abstraction or analysis applied to language data.
- In physics, level can refer to the energy state of an atom or particle.
- In chemistry, level can refer to the concentration of a substance within a solution or mixture.

CO-1 Get an understanding of the use of different analytical instruments. K3

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are commonly used in scientific research, quality control, and medical diagnosis. Some of the most commonly used analytical instruments include:

1. **Spectrophotometers:** These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.
2. **Chromatographs:** These instruments separate and analyze the components of a mixture. They are commonly used to identify and quantify the components of a sample.
3. **Mass Spectrometers:** These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.
4. **Electrochemical Analyzers:** These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.
5. **Thermal Analyzers:** These instruments measure the thermal properties of a sample. They are commonly used to determine the melting point, boiling point, and heat capacity of a substance.

Understanding the use of these different analytical instruments is important for conducting accurate and reliable experiments and analyses. Each instrument has its own strengths and limitations, and selecting the appropriate instrument for a particular task is crucial for obtaining accurate results.

CO-2 Measure the molecular / system properties such as surface tension

Surface tension is a physical property of liquids that results from the cohesive forces between the molecules at the surface of a liquid. It is defined as the force acting per unit length on the surface of a liquid, and is typically measured in units of N/m or dyn/cm.

There are several methods for measuring surface tension, including the following:

1. **Maximum Bubble Pressure Method:** This method involves measuring the maximum pressure required to force a gas bubble through a small orifice submerged in the liquid. The surface tension is calculated from the maximum pressure and the radius of the orifice.
2. **Drop Weight Method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is calculated from the weight of the droplet and the radius of the tube.
3. **Wilhelmy Plate Method:** This method involves immersing a thin, flat plate into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is calculated from the force and the perimeter of the plate.
4. **Du Nouy Ring Method:** This method involves immersing a thin, circular ring into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is calculated from the force and the circumference of the ring.

These are some of the common methods used to measure surface tension. Each method has its own advantages and limitations, and the choice of method may depend on the specific application and the accuracy required. It is important to carefully calibrate and standardize the measurement equipment to ensure accurate and reliable results.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

1. **Viscosity** is a measure of a fluid's resistance to flow. It is determined by the internal friction of the fluid, which is caused by the cohesive forces between the molecules. The viscosity of formation water is a function of pressure, temperature, and dissolved solids. In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature. Dissolved gas in the formation water at reservoir conditions generally results in a negligible effect on water viscosity .
2. **Conductance of solution** is the ability of a solution to conduct an electric current. The more ions present in the solution, the higher the

conductivity. High-quality deionized water has a conductivity of about 0.05 S/cm at 25 °C, typical drinking water is in the range of 200–800 S/cm, while seawater is about 50 mS/cm (or 0.05 S/cm) .

3. **Chloride content in water** can affect its conductivity. Chloride ions are approximately 1.9 percent of the mass of seawater. The more chloride ions present in the water, the higher the conductivity. In general, the conductivity of waterways in the United States varies from 50 to 1500 $\mu\text{mhos/cm}$, and inland freshwater lake studies reveal a conductivity of 150 to 500 $\mu\text{mhos/cm}$.
4. **Iron content in water** is not mentioned in the search results. Further information is needed to provide a comprehensive response on this topic.

K3

- K3 is a term that can refer to several different things, depending on the context.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a visa category for spouses of U.S. citizens.
- In mathematics, K3 could refer to a complete graph with three vertices.
- In music, K3 is a Belgian-Dutch girl group.
- In transportation, K3 may refer to a class of steam locomotives, a model of Kia car, or a type of kayak.

CO-3 Measure the hardness and alkalinity of the water. K3

Hardness and alkalinity are two important parameters to measure the quality of water. Hardness refers to the concentration of calcium and magnesium ions in water, while alkalinity refers to the ability of water to neutralize acids.

1. **Measuring Hardness:** Hardness can be measured using a titration method, where a solution of ethylenediaminetetraacetic acid (EDTA) is used to titrate the water sample. The endpoint of the titration is indicated by a color change, which occurs when all the calcium and magnesium ions have reacted with the EDTA. The hardness of the water can then be calculated based on the volume of EDTA used.
2. **Measuring Alkalinity:** Alkalinity can be measured using a titration method, where a solution of a strong acid, such as hydrochloric acid (HCl), is used to titrate the water sample. The endpoint of the titration is indicated by a pH meter or a color change, which occurs when all the acid-neutralizing compounds in the water have reacted with the acid. The alkalinity of the water can then be calculated based on the volume of acid used.

It is important to measure the hardness and alkalinity of water to ensure that it is suitable for its intended use, such as for drinking, irrigation, or industrial processes. High levels of hardness can cause scaling in pipes and appliances, while low levels of alkalinity can make water corrosive. By measuring these parameters, appropriate treatment methods can be implemented to improve the quality of the water.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that is also known as carbolic acid. It is a weak acid that mostly forms phenoxide ions by dropping one positive hydrogen ion (H^+) from the hydroxyl group. There are several methods for the preparation of phenol, including:

1. From Haloarenes: Chlorobenzene is an example of a haloarene which is formed by the monosubstitution of the benzene ring. When chlorobenzene is fused with sodium hydroxide at 623K and 320 atm, sodium phenoxide is produced .
2. From Benzene Sulphonic Acid: At 573 – 623K, sodium phenoxide is produced by fusing the sodium salt of benzene sulphonic acid with sodium hydroxide. Sodium phenoxide is converted to phenol when it is treated with dilute acid or carbon dioxide.
3. From Cumene: Phenol can be prepared in large quantities by the air oxidation of cumene.
4. From Diazonium Salts: Phenol can be prepared in small amounts by the hydrolysis of diazonium salts.
5. From Coal Tar: Phenol is extracted from the middle oil fraction for commercial purposes at 443 – 503K.
6. Raschig's Process: The reaction of benzene, hydrogen chloride and air at 523K produces phenol.

Formaldehyde, Urea Formaldehyde Resin, Adipic Acid and Paracetamol

Formaldehyde

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- It is used in pressed-wood products, such as particleboard, plywood, and fiberboard; glues and adhesives; permanent-press fabrics; paper product coatings; and certain insulation materials.

Urea Formaldehyde Resin

- Urea-formaldehyde (UF), also known as urea-methanal, is a nontransparent thermosetting resin or polymer.
- It is produced from urea and formaldehyde.
- These resins are used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- Using formaldehyde and urea as raw materials, a stable urea-formaldehyde resin (UF) is synthesized by the “alkali-acid-alkali” method .
- Unlike most thermosetting resins, UF often shows the appearance of crystal domains .

Adipic Acid

- Adipic acid dihydrazide can be used as a modifier to reduce the free formaldehyde in urea-formaldehyde (UF) resin.
- The influence of the addition amount and addition stage of modifiers on the physicochemical property and free formaldehyde content of UF resin can be tested.

Paracetamol

K3

K3 is a term that can refer to several different things. Without more context, it is difficult to determine the specific meaning of K3 in this case. Some possible meanings of K3 include:

1. K3 is a mountain in the Karakoram range, also known as Broad Peak.
2. K3 is a visa category for spouses of U.S. citizens.
3. K3 is a type of potassium channel.
4. K3 is a Belgian-Dutch girl group.
5. K3 is a type of tax form used in some countries.

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by the symbol k , is a measure of the speed of a chemical reaction. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers. The value of k is dependent on the temperature, pressure, and the nature of the reactants.

To estimate the rate constant of a reaction, the following steps can be followed:

1. Conduct a series of experiments to measure the rate of the reaction at different concentrations of the reactants.

2. Plot the data on a graph with the rate of the reaction on the y-axis and the concentration of the reactants on the x-axis.
3. Determine the order of the reaction with respect to each reactant by analyzing the shape of the graph.
4. Use the integrated rate law for the reaction to calculate the value of k from the experimental data.

It is important to note that the value of k is only valid for the specific conditions under which the experiments were conducted. If the temperature, pressure, or nature of the reactants changes, the value of k will also change.

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- Page 29 is the 29th page of a 40-page document.
- It is located after page 28 and before page 30.
- The content of page 29 is dependent on the document it is a part of.
- It may contain text, images, tables, or other types of information.
- The information on page 29 may be related to the information on the previous and following pages.
- It is important to read and understand the content of page 29 in the context of the entire document.
- Page 29 may contain important information for exams or other purposes.

ENGINEERING PHYSICS LAB KCS

Engineering Physics Lab KCS is a laboratory course that is designed to provide students with hands-on experience in the application of physics principles to engineering problems. The course typically covers topics such as mechanics, thermodynamics, electromagnetism, and optics.

Some of the key objectives of the Engineering Physics Lab KCS course include:

- To provide students with practical experience in the application of physics principles to engineering problems.
- To develop students' skills in experimental design, data analysis, and scientific communication.
- To enhance students' understanding of the fundamental principles of physics and their relevance to engineering.

The course typically includes a series of laboratory experiments that are designed to reinforce the concepts covered in the lectures. These experiments may include: - Measurement of physical quantities such as mass, length, time, and temperature. - Study of the properties of materials, including elasticity, thermal conductivity, and electrical conductivity. - Investigation of the behavior of mechanical systems, including simple harmonic motion, resonance, and damping. - Analysis of the behavior of electrical circuits, including Ohm's law, Kirchhoff's laws, and the behavior of capacitors and inductors. - Study of the properties of light, including reflection, refraction, and interference.

The Engineering Physics Lab KCS course is typically assessed through a combination of laboratory reports, quizzes, and exams. Students are expected to demonstrate their understanding of the concepts covered in the course, as well as their ability to apply these concepts to the analysis of experimental data.

Course Objectives

- To provide students with a comprehensive understanding of the subject matter.
- To develop critical thinking and analytical skills.
- To enhance problem-solving abilities.
- To encourage independent learning and self-motivation.
- To promote effective communication and collaboration.
- To prepare students for further academic or professional pursuits.
- To foster an appreciation for the subject and its real-world applications.
- To instill a sense of ethical responsibility and social awareness.

Fundamental Concepts of Analytical Instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are commonly used in scientific research, medical diagnosis, and industrial processes.

Here are some fundamental concepts of analytical instruments:

1. **Accuracy:** The degree to which the measurement result agrees with the true value of the quantity being measured.
2. **Precision:** The degree to which repeated measurements of the same quantity agree with each other.
3. **Sensitivity:** The smallest change in the quantity being measured that can be detected by the instrument.
4. **Selectivity:** The ability of the instrument to distinguish between different components in a mixture.
5. **Calibration:** The process of adjusting the instrument to ensure that its measurements are accurate.
6. **Signal-to-Noise Ratio:** The ratio of the signal strength to the background noise level. A high signal-to-noise ratio indicates a clear and reliable measurement.
7. **Resolution:** The smallest difference between two measurements that can be distinguished by the instrument.
8. **Dynamic Range:** The range of values over which the instrument can accurately measure.

These concepts are important to understand when using analytical instruments to ensure accurate and reliable measurements. It is also important to properly maintain and calibrate the instruments to ensure their continued accuracy.

Analysis of Chloride Content, Hardness, and Alkalinity

1. **Chloride Content:** Chloride content refers to the amount of chloride ions present in a sample of water. Chloride ions are negatively charged particles that can be found in many types of salts, including sodium chloride (common table salt). The presence of chloride ions in water can indicate contamination from sources such as seawater intrusion, industrial waste, or road salt.
2. **Hardness:** Water hardness refers to the concentration of dissolved minerals, primarily calcium and magnesium, in a sample of water. These minerals can cause scaling and buildup in pipes and appliances, and can also affect the taste and feel of the water. Hardness is typically measured in grains per gallon (gpg) or milligrams per liter (mg/L).
3. **Alkalinity:** Alkalinity refers to the ability of water to neutralize acids. It is a measure of the buffering capacity of water, or its ability to resist changes in pH. Alkalinity is primarily influenced by the presence of bicarbonate, carbonate, and hydroxide ions in water. It is typically measured in milligrams per liter (mg/L) as calcium carbonate (CaCO_3).

These three parameters are important in the analysis of water quality and can provide valuable information about the sources and potential impacts of contamination. Understanding these parameters can help in the management and treatment of water resources.

Water

Water is a transparent, tasteless, odorless, and nearly colorless chemical substance, which is the main constituent of Earth's streams, lakes, and oceans, and the fluids of most living organisms. It is vital for all known forms of life, even though it provides no calories or organic nutrients.

- Water is a liquid at standard temperature and pressure.
- Water covers 71% of the Earth's surface.
- Water is the most abundant compound on Earth's surface.
- Water is essential for the survival of all known forms of life.
- Water is a polar molecule, meaning it has a partial positive charge near the hydrogen atoms and a partial negative charge near the oxygen atom.
- Water has a high heat capacity, meaning it can absorb a lot of heat before its temperature rises.
- Water has a high heat of vaporization, meaning it takes a lot of heat to turn it into vapor.
- Water is a good solvent, meaning it can dissolve many substances.
- Water is transparent, allowing sunlight to penetrate and support photosynthesis in aquatic plants.

- Water expands when it freezes, becoming less dense than its liquid form. This causes ice to float on water.

3. Measure of pH, Surface Tension and Viscosity

pH

- pH is a measure of the acidity or basicity of a solution.
- It is defined as the negative logarithm of the hydrogen ion concentration in a solution.
- The pH scale ranges from 0 to 14, with 7 being neutral.
- A pH value less than 7 indicates an acidic solution, while a pH value greater than 7 indicates a basic solution.

Surface Tension

- Surface tension is the property of a liquid that allows it to resist an external force.
- It is caused by the cohesive forces between the molecules of the liquid.
- Surface tension is measured in units of force per unit length, typically in millinewtons per meter (mN/m).
- The surface tension of a liquid can be affected by various factors, including temperature, the presence of surfactants, and the nature of the liquid itself.

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- It is defined as the ratio of the shear stress to the shear rate in a fluid.
- Viscosity is typically measured in units of pascal-seconds ($\text{Pa} \cdot \text{s}$).
- The viscosity of a fluid can be affected by various factors, including temperature, pressure, and the presence of impurities.

Liquid

- A liquid is one of the four fundamental states of matter, along with solid, gas, and plasma.
- It is characterized by its ability to conform to the shape of its container, while retaining a constant volume.
- Liquids have a definite volume, but no definite shape.
- The particles in a liquid are close together and are in constant motion, allowing them to slide past one another.
- This gives liquids the ability to flow and take the shape of their container.
- The attractive forces between the particles in a liquid are strong enough to keep the particles close together, but not strong enough to keep them in a fixed position.

- This results in a liquid having a fixed volume, but not a fixed shape.
- Liquids have a much higher density than gases, but a lower density than solids.
- The density of a liquid can change with temperature and pressure.
- Liquids can be compressed, but not as easily as gases.
- The surface of a liquid can exhibit surface tension, which is the result of the cohesive forces between the liquid particles.
- Liquids can also exhibit capillary action, which is the ability of a liquid to flow against gravity in a narrow space, such as a thin tube.
- The study of liquids and their properties is known as fluid mechanics.

Preparation of Different Resins

Resins are solid or highly viscous substances that are typically convertible into polymers. They are used in a variety of applications, including adhesives, coatings, and plastics. Here are some points to consider when preparing different resins:

1. **Type of resin:** There are several types of resins, including natural resins (such as rosin and shellac) and synthetic resins (such as epoxy and polyurethane). The type of resin used will depend on the desired properties and application.
2. **Mixing:** Many resins require mixing with a hardener or catalyst before use. It is important to follow the manufacturer's instructions for the correct ratio and mixing procedure.
3. **Curing:** Resins typically require a certain amount of time to cure or harden. The curing time and conditions (such as temperature and humidity) will vary depending on the type of resin and the application.
4. **Safety:** Resins can be hazardous to work with, so it is important to follow safety guidelines and wear appropriate protective equipment, such as gloves and eye protection.
5. **Storage:** Resins should be stored in a cool, dry place, away from direct sunlight and heat sources. Some resins have a limited shelf life and may degrade over time if not stored properly.

By following these guidelines, students can learn how to prepare different resins for various applications. It is important to always follow the manufacturer's instructions and safety guidelines when working with resins.

Synthesis of Adipic Acid

Adipic acid is an important organic compound that is used in the production of nylon, polyurethane, and other industrial products. Here are some key points

to understand about the synthesis of adipic acid:

1. Adipic acid is a dicarboxylic acid with the chemical formula $C_6H_{10}O_4$.
2. It can be synthesized from a variety of starting materials, including cyclohexanol, cyclohexanone, and benzene.
3. One common method for synthesizing adipic acid is through the oxidation of cyclohexanol or cyclohexanone using nitric acid.
4. Another method involves the oxidation of benzene using a mixture of nitric and sulfuric acids.
5. The resulting adipic acid can be purified through recrystallization or other separation techniques.

These are some of the key points to understand about the synthesis of adipic acid. It is an important industrial chemical with a variety of uses, and its synthesis involves several steps and chemical reactions.

Acid and Paracetamol by Conventional and Green Route

- Paracetamol, also known as acetaminophen, is a common pain reliever and fever reducer.
- The conventional synthesis of paracetamol involves the nitration of phenol to produce p-nitrophenol, which is then reduced to p-aminophenol and acetylated to produce paracetamol.
- An alternative route for the synthesis of paracetamol using principles of green chemistry has been developed. This route involves solvent-free synthesis combined with microwave-energy irradiation .
- In 2014, Joncour et al. established a green approach involving direct amidation of hydroquinone to obtain paracetamol with maximum conversion. This approach eliminates the problems associated with the earlier techniques such as the extra step involved, unwanted ortho-isomerization, and generation of the sulfate salt .
- Green chemistry aims to reduce the environmental impact of chemical processes by minimizing the use of hazardous substances, reducing waste, and increasing energy efficiency.

List of Experiments

1. **The Milgram Experiment:** A series of social psychology experiments conducted by Stanley Milgram in the 1960s, which measured the willingness of study participants to obey an authority figure who instructed them to perform acts that conflicted with their personal conscience.
2. **The Stanford Prison Experiment:** A social psychology study conducted by Philip Zimbardo in 1971, which investigated the psychological

effects of perceived power, focusing on the struggle between prisoners and prison officers.

3. **The Asch Conformity Experiment:** A series of studies conducted by Solomon Asch in the 1950s, which demonstrated the power of conformity in groups and the influence of the majority on individual behavior.
4. **The Pavlov's Dog Experiment:** A study conducted by Ivan Pavlov in the 1890s, which investigated classical conditioning and the association of stimuli with reflex responses.
5. **The Marshmallow Test:** A study conducted by Walter Mischel in the 1960s and 1970s, which investigated delayed gratification and the ability of children to resist the temptation of an immediate reward in favor of a larger reward later on.

Calibration of Analytical Equipment and Apparatus

Calibration is the process of verifying the accuracy of an instrument or apparatus by comparing its measurements with a known standard. This is important for ensuring that the results obtained from the instrument are accurate and reliable.

1. Calibration should be performed regularly to ensure that the instrument is functioning properly and providing accurate results.
2. The frequency of calibration depends on the instrument and its usage, but it is generally recommended to calibrate at least once a year.
3. Calibration can be performed by the manufacturer, a third-party calibration service, or in-house using a calibration standard.
4. Calibration standards are materials or devices with known properties that are used to verify the accuracy of an instrument.
5. The calibration process involves comparing the measurements obtained from the instrument with the known values of the calibration standard.
6. If the instrument is found to be inaccurate, it can be adjusted or repaired to bring it back into specification.
7. Documentation of the calibration process, including the date, results, and any adjustments made, is important for maintaining a record of the instrument's performance.

Calibration is an essential part of maintaining the accuracy and reliability of analytical equipment and apparatus. It is important to follow the manufacturer's recommendations for calibration frequency and to keep accurate records of the calibration process.

Determination of Hardness of Water Sample by EDTA Method

1. **Introduction:** Hardness of water is a measure of the concentration of calcium and magnesium ions in water. These ions form insoluble compounds with soap, causing the formation of scum and reducing the effectiveness of the soap. The hardness of water can be determined by titration with a chelating agent, such as ethylenediaminetetraacetic acid (EDTA).
2. **Principle:** The EDTA method for determining the hardness of water is based on the complexometric titration of calcium and magnesium ions with EDTA. In this method, a solution of EDTA is added to a water sample until all the calcium and magnesium ions have formed complexes with the EDTA. The end point of the titration is indicated by a color change of a metal ion indicator, such as Eriochrome Black T.
3. **Procedure:** To determine the hardness of a water sample by the EDTA method, the following steps are followed:
 1. A known volume of the water sample is pipetted into a conical flask.
 2. A few drops of buffer solution are added to the flask to adjust the pH of the solution.
 3. A few drops of Eriochrome Black T indicator are added to the flask. The solution turns wine red in color.
 4. The solution is titrated with a standard solution of EDTA until the color changes from wine red to blue.
 5. The volume of EDTA solution used is recorded.
4. **Calculations:** The hardness of the water sample can be calculated from the volume of EDTA solution used in the titration. The concentration of calcium and magnesium ions in the water sample can be calculated from the volume and molarity of the EDTA solution used. The hardness of the water sample is expressed in terms of milligrams of calcium carbonate per liter of water (mg CaCO_3/L).
5. **Conclusion:** The EDTA method is a simple and accurate method for determining the hardness of water. It is widely used in water treatment plants and in the analysis of water samples.

Determination of Alkalinity of Water Sample

1. Alkalinity is a measure of the ability of water to neutralize acids. It is determined by the concentration of bicarbonate, carbonate, and hydroxide ions present in the water sample.
2. Alkalinity is important because it affects the pH of the water and can influence the solubility of certain substances.
3. To determine the alkalinity of a water sample, a titration method is commonly used. This involves adding a known amount of acid to the water sample until a pH endpoint is reached.
4. The amount of acid required to reach the endpoint is used to calculate the

alkalinity of the water sample.

5. The endpoint of the titration is typically determined using a pH indicator or a pH meter.
6. The results of the alkalinity test are usually reported in milligrams per liter (mg/L) as calcium carbonate (CaCO_3).
7. Alkalinity can also be determined using other methods such as colorimetric or ion chromatography, but these methods are less common.
8. Regular monitoring of the alkalinity of water sources is important for maintaining water quality and ensuring the health of aquatic ecosystems.

Determination of pH by Titrimetric Method

1. Titrimetric method is a common laboratory technique used to determine the pH of a solution.
2. It involves the addition of a titrant, which is a solution of known concentration, to the solution being tested.
3. The titrant is added until the reaction between the two solutions is complete, which is indicated by a change in color of an indicator or by a change in the electrical potential of the solution.
4. The volume of titrant added is used to calculate the concentration of the solution being tested, and from this, the pH can be determined.
5. Common titrants used in pH determination include strong acids or bases, such as hydrochloric acid or sodium hydroxide.
6. The choice of indicator used depends on the pH range of the solution being tested. For example, phenolphthalein is commonly used as an indicator for solutions with a pH range of 8.2-10.0, while methyl orange is used for solutions with a pH range of 3.1-4.4.
7. The accuracy of the titrimetric method depends on the precision of the measurements and the purity of the reagents used.
8. It is important to calibrate the equipment and standardize the titrant before performing the titration to ensure accurate results.

Determination of Surface Tension of Given Liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the molecules at the surface. It is defined as the force acting per unit length on the surface of a liquid. There are several methods to determine the surface tension of a given liquid, including:

1. **Capillary rise method:** This method involves measuring the height of a liquid in a capillary tube. The surface tension is calculated using the formula: $\gamma = \frac{h \rho g r}{2 \cos \theta}$, where h is the height of the liquid, ρ is the density of the liquid, g is the acceleration due to gravity, and θ is the contact angle between the liquid and the capillary tube.

2. **Maximum bubble pressure method:** This method involves measuring the pressure required to form a bubble at the end of a submerged tube. The surface tension is calculated using the formula: $\gamma = (4P_m)/d$, where P_m is the maximum pressure and d is the diameter of the tube.
3. **Drop weight method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a tube. The surface tension is calculated using the formula: $\gamma = mg/d$, where m is the mass of the droplet, g is the acceleration due to gravity, and d is the diameter of the tube.
4. **Wilhelmy plate method:** This method involves immersing a thin, flat plate into a liquid and measuring the force required to detach it. The surface tension is calculated using the formula: $\gamma = F/2l$, where F is the force and l is the perimeter of the plate.
5. **Du Nouy ring method:** This method involves immersing a ring into a liquid and measuring the force required to detach it. The surface tension is calculated using the formula: $\gamma = (F - F_c)/4R$, where F is the force, F_c is the correction factor, and R is the radius of the ring.

Each of these methods has its own advantages and limitations, and the choice of method depends on the specific requirements of the experiment. It is important to carefully calibrate the instruments and follow the experimental procedure to obtain accurate results.

Determination of Viscosity of a given liquid by viscometer

1. Viscosity is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate in a fluid.
2. A viscometer is an instrument used to measure the viscosity of a fluid. There are several types of viscometers, including capillary, rotational, and falling ball viscometers.
3. To determine the viscosity of a given liquid using a viscometer, the liquid is placed in the instrument and its flow rate is measured.
4. The flow rate is then used to calculate the viscosity of the liquid using the appropriate formula for the type of viscometer being used.
5. The viscosity of the liquid can be affected by factors such as temperature and pressure, so these should be controlled during the experiment.
6. The results of the experiment can be used to compare the viscosities of different liquids or to determine how the viscosity of a liquid changes under different conditions.

Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. Its chemical formula is $(\text{NH}_4)_2\text{Fe}(\text{SO}_4) \cdot 2.6\text{H}_2\text{O}$.
3. It is commonly used as a standard solution in redox titrations due to its stability and the ease with which it can be prepared.
4. The strength of a ferrous ammonium sulfate solution can be determined using an external indicator.
5. An external indicator is a substance that changes color at the endpoint of a titration.
6. The most commonly used external indicator for the determination of the strength of ferrous ammonium sulfate is potassium permanganate (KMnO_4).
7. Potassium permanganate is a strong oxidizing agent and its solution is purple in color.
8. During the titration, the purple color of the potassium permanganate solution fades as it is reduced to colorless manganese ions.
9. The endpoint of the titration is reached when a faint pink color persists, indicating that all the ferrous ions have been oxidized to ferric ions.
10. The strength of the ferrous ammonium sulfate solution can then be calculated using the volume of potassium permanganate solution used and the concentration of the potassium permanganate solution.

Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is an oxidizing agent that can be used to determine the strength of a substance.
2. An internal indicator is used to determine the end point of the titration.
3. The internal indicator changes color when the reaction is complete, indicating that the titration is finished.
4. The strength of the potassium dichromate can then be calculated using the volume of the titrant and the concentration of the substance being titrated.
5. This method is commonly used in analytical chemistry to determine the strength of a substance.

Determination of Available Chlorine in Bleaching Powder

1. **Aim:** To determine the available chlorine in the given sample of bleaching powder by the iodometric method.
2. **Principle:** Bleaching powder is commonly used as a disinfectant. The chlorine present in the bleaching powder gets reduced with time.
3. **Method:** The available chlorine present in the bleaching powder sample is determined iodometrically by treating its solution with an excess of potassium iodide solution in the acidic medium. The liberated iodine (I_2) is treated with sodium thiosulphate ($Na_2S_2O_3$) solution using freshly prepared starch solution as an indicator to be added near the endpoint.
4. **Calculation:** The chlorine liberated is called available chlorine, and the calculation of available chlorine is called iodometry. Take a W (gm) sample of bleaching powder; and when bleaching powder is treated with dilute acid or water, it liberates chlorine gas.

Determination of Chloride Content in Water Sample

1. Chloride is one of the major anions found in water and wastewater.
2. The chloride content of water can be determined by titration with silver nitrate solution, using potassium chromate as an indicator.
3. The reaction between silver nitrate and chloride ions produces a white precipitate of silver chloride.
4. The end point of the titration is indicated by the formation of a red-brown color, due to the reaction between excess silver ions and chromate ions.
5. The concentration of chloride ions in the water sample can be calculated from the volume of silver nitrate solution used in the titration.
6. The presence of other anions, such as bromide and iodide, can interfere with the determination of chloride content.
7. To overcome this interference, the water sample can be treated with nitric acid to convert these anions to their respective acids, which do not interfere with the titration.
8. The determination of chloride content in water is important for monitoring the quality of drinking water and wastewater, as well as for industrial and agricultural applications.
9. High levels of chloride in water can have negative impacts on aquatic life and can also affect the taste of drinking water.
10. Regular monitoring of chloride levels in water sources is necessary to ensure the safety and quality of water for human consumption and for the environment.

Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resins are synthetic polymers obtained by the reaction of phenol or substituted phenol with formaldehyde. They were the first commercial synthetic resins (plastics) and are used as the basis for Bakelite.

- **Preparation:** A step-growth polymerization reaction that can be either acid- or base-catalyzed is used for the preparation of Phenol-formaldehyde resins. Since formaldehyde exists as a dynamic equilibrium of methylene glycol oligomers for the most part in the solution, it is a reactive derivative of methane.
- **Control PF resin:** Control PF resin with a formaldehyde to phenol (F/P) molar ratio of 2.0 can be synthesized with phenol charged once and formaldehyde charged twice. NaOH (40% solution) can be divided into two portions and charged with formaldehyde respectively.
- **Heating:** Phenol, formaldehyde, water, and catalyst are mixed in the desired amount, depending on the resin to be formed, and are then heated. The first part of the reaction, at around 70 °C, forms a thick reddish-brown tacky material, which is rich in hydroxymethyl and benzylic ether groups.

Preparation of Urea Formaldehyde (UF) Resin

Urea Formaldehyde (UF) resin is a type of thermosetting synthetic polymer that is commonly used in the production of adhesives, finishes, and molded objects. Here are the steps involved in the preparation of UF resin:

1. **Reacting Urea and Formaldehyde:** The first step in the preparation of UF resin is the reaction of urea and formaldehyde in an aqueous solution. The molar ratio of urea to formaldehyde is typically between 1:1.5 and 1:2.5. The reaction is carried out at a pH of 7-8 and a temperature of 70-90°C.
2. **Condensation:** The next step is the condensation of the reaction mixture. This is done by heating the mixture to a temperature of 90-100°C and maintaining it at that temperature for 1-3 hours. During this time, the water content of the mixture is reduced, and the condensation reaction takes place, forming the UF resin.
3. **Cooling and Neutralization:** After the condensation reaction is complete, the mixture is cooled to room temperature. The pH of the mixture is then adjusted to 7-8 using an acid or a base.
4. **Filtration and Drying:** The final step in the preparation of UF resin is the filtration and drying of the resin. The resin is filtered to remove any impurities and then dried to remove any remaining water.

These are the basic steps involved in the preparation of UF resin. The exact conditions and parameters of the reaction may vary depending on the desired

properties of the final product. It is important to carefully control the reaction conditions to ensure the production of a high-quality UF resin.

Preparation of Adipic acid / Paracetamol

Adipic acid

Adipic acid is an organic compound with the formula $(\text{CH}_2)_4(\text{COOH})_2$. It is the most important dicarboxylic acid from an industrial perspective, with about 2.5 billion kilograms of this white crystalline powder being produced annually, mainly as a precursor for the production of nylon .

Preparation

Adipic acid is produced from a mixture of cyclohexanone and cyclohexanol called KA oil, the abbreviation of ketone-alcohol oil. The KA oil is oxidized with nitric acid to give adipic acid, via a multistep pathway .

Paracetamol

Paracetamol is an effective antipyretic and analgesic .

Preparation

Paracetamol is prepared from p-aminophenol by acetylating it with acetic anhydride in the presence of 3-4 drops of concentrated sulphuric acid as a catalyst .

Determination of Cell Conductance of a Solution

1. Cell conductance is a measure of a solution's ability to conduct electricity.
2. It is determined by measuring the electrical conductivity of the solution using a conductivity meter.
3. The conductivity meter consists of a cell with two electrodes and a circuit to measure the current passing through the solution.
4. The cell constant is determined by measuring the conductivity of a solution of known conductivity.
5. The cell conductance is then calculated by dividing the measured conductivity by the cell constant.
6. The cell conductance is affected by the concentration and mobility of the ions in the solution, as well as the temperature of the solution.
7. The cell conductance can be used to determine the concentration of ions in a solution, or to monitor changes in the concentration of ions over time.

Determination of Rate constant of hydrolysis of esters

1. The hydrolysis reaction of an ester in pure water is a slow reaction.
2. When a mineral acid like hydrochloric acid is added, the rate of the reaction is enhanced since the H^+ ions from the mineral acid act as the catalyst.
3. The hydrolysis of esters is catalyzed by either an acid or a base.
4. Acidic hydrolysis is simply the reverse of esterification.
5. The ester is heated with a large excess of water containing a strong-acid catalyst.
6. Like esterification, the reaction is reversible and does not go to completion.
7. The rate constant for the hydrolysis reaction of an ester by dilute acid is $0.6931 \times 10^{-3} \text{ s}^{-1}$.
8. The time required to change the concentration of ester from 0.04 M to 0.01 M is 6931 sec.
9. The rate constant after calculation from the graphs was approximately 0.003 min^{-1} for the 1ml and 2ml of ethyl acetate.
10. When ethyl ethanoate is heated under reflux with a dilute acid such as dilute hydrochloric acid or dilute sulfuric acid, the ester reacts with the water present to produce ethanoic acid and ethanol.

Element Detection and Identification of Functional Groups in Organic Compounds

1. **Elemental Analysis:** Elemental analysis is a process where a sample of an organic compound is analyzed for its elemental composition. This can be done through various methods such as combustion analysis, mass spectrometry, and X-ray fluorescence spectroscopy.
2. **Detection of Carbon and Hydrogen:** Carbon and hydrogen are the two most common elements in organic compounds. They can be detected through combustion analysis, where the sample is burned in the presence of oxygen to produce carbon dioxide and water. The amount of carbon dioxide and water produced can be used to calculate the amount of carbon and hydrogen in the sample.
3. **Detection of Nitrogen:** Nitrogen can be detected through the Dumas method or the Kjeldahl method. In the Dumas method, the sample is heated in the presence of copper oxide to produce nitrogen gas, which is then measured. In the Kjeldahl method, the sample is digested with sulfuric acid to convert nitrogen into ammonium sulfate, which is then measured.
4. **Detection of Halogens:** Halogens can be detected through the Carius method, where the sample is heated in the presence of silver nitrate to

produce silver halides, which can be weighed to determine the amount of halogens in the sample.

5. **Detection of Sulfur:** Sulfur can be detected through the Carius method, where the sample is heated in the presence of barium nitrate to produce barium sulfate, which can be weighed to determine the amount of sulfur in the sample.
6. **Identification of Functional Groups:** Functional groups are specific groups of atoms within an organic molecule that are responsible for its chemical properties. They can be identified through various methods such as infrared spectroscopy, nuclear magnetic resonance spectroscopy, and mass spectrometry.
7. **Infrared Spectroscopy:** Infrared spectroscopy is a technique that measures the absorption of infrared radiation by a molecule. Different functional groups absorb infrared radiation at different wavelengths, allowing for their identification.
8. **Nuclear Magnetic Resonance Spectroscopy:** Nuclear magnetic resonance spectroscopy is a technique that measures the magnetic properties of atomic nuclei. Different functional groups have different magnetic properties, allowing for their identification.
9. **Mass Spectrometry:** Mass spectrometry is a technique that measures the mass-to-charge ratio of ions. Different functional groups produce characteristic fragmentation patterns, allowing for their identification.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose any 10 experiments from the list provided.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read and understand the instructions provided by the instructor.
- Students should also be prepared for any changes to the list of experiments.

Course Outcomes

Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course. These outcomes serve

as a guide for instructors in designing course content and assessments, and for students in understanding the goals of the course.

Some key points to consider when developing course outcomes include:

1. Outcomes should be specific, measurable, and achievable within the scope of the course.
2. Outcomes should be aligned with the overall goals of the program or curriculum.
3. Outcomes should be written in clear, concise language that is easily understood by students.
4. Outcomes should be reviewed and updated regularly to ensure they remain relevant and accurate.

By clearly defining course outcomes, instructors can ensure that their course content and assessments are aligned with the goals of the course, and students can better understand what is expected of them and what they will gain from the course. This can lead to more effective teaching and learning, and ultimately, to better student outcomes.

Upon completion of the course the student should be able to:

- Demonstrate an understanding of the fundamental concepts and principles of the subject matter.
- Apply the knowledge and skills acquired in the course to solve problems and make informed decisions.
- Communicate effectively using the language and terminology of the subject matter.
- Work collaboratively with others to achieve common goals.
- Demonstrate the ability to think critically and creatively.
- Show an awareness of the ethical, social, and environmental implications of the subject matter.
- Engage in self-directed learning and take responsibility for their own learning.
- Demonstrate the ability to use technology effectively in the learning process.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course. These outcomes are typically aligned with the course objectives and are used to assess the effectiveness of the course in achieving its goals.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is often used to design course outcomes and assessments. The taxonomy consists of six levels, arranged in a hierarchy from lower-order thinking skills to higher-order thinking skills:

1. **Remembering:** The ability to recall or recognize information.
2. **Understanding:** The ability to comprehend the meaning of information.
3. **Applying:** The ability to use information in a new situation.
4. **Analyzing:** The ability to break down information into parts and understand their relationships.
5. **Evaluating:** The ability to make judgments based on criteria and standards.
6. **Creating:** The ability to combine elements to form a new whole.

Course outcomes can be written using verbs from Bloom's Taxonomy to indicate the level of thinking required. For example, a course outcome for a history course might be: "Students will be able to analyze primary source documents to construct an argument about a historical event." This outcome uses the verb "analyze" from the fourth level of Bloom's Taxonomy.

In summary, course outcomes are statements that describe what students should be able to do by the end of a course, and Bloom's Taxonomy is a framework that can be used to design and assess these outcomes. By using verbs from Bloom's Taxonomy, course outcomes can be written to reflect the level of thinking required of students.

Level

- A level is a tool used to determine if a surface is horizontal (level) or vertical (plumb).
- Levels can be used in construction, carpentry, surveying, and many other applications.
- There are several types of levels, including spirit levels, laser levels, and water levels.
- Spirit levels use a liquid-filled vial with an air bubble to indicate levelness. The bubble will be centered between two lines when the surface is level.
- Laser levels project a laser beam to create a straight line, which can be used to determine levelness.
- Water levels use the principle that water will always find its own level. A clear tube filled with water can be used to determine if two points are at the same elevation.
- Levels can vary in size and accuracy, with larger and more precise levels being used for more demanding applications.
- It is important to use a level when installing or building structures to ensure that they are straight and level, which can affect their stability and appearance.

CO-1 Get an understanding of the use of different analytical instruments. K3

Analytical instruments are used to measure physical, chemical, and biological properties of samples. These instruments are used in a wide range of fields, including research, quality control, and medical diagnostics. Some common analytical instruments include:

1. **Spectrophotometers:** These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a sample.
2. **Chromatographs:** These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.
3. **Mass spectrometers:** These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify unknown compounds and to determine the elemental composition of a sample.
4. **Electrochemical instruments:** These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.
5. **Microscopes:** These instruments magnify small objects or samples. They are commonly used to observe the structure of cells and tissues.

Each of these instruments has its own unique capabilities and limitations. Understanding how to use these instruments and interpret their results is essential for conducting accurate and reliable analyses.

CO2: Measuring Molecular/System Properties such as Surface Tension

Carbon dioxide (CO₂) is a colorless, odorless gas that is naturally present in the Earth's atmosphere. It is an important greenhouse gas that helps regulate the temperature of the planet. However, human activities such as burning fossil fuels and deforestation have increased the concentration of CO₂ in the atmosphere, contributing to global warming.

Surface tension is a property of liquids that arises from the cohesive forces between molecules at the surface of the liquid. It is the force that allows insects to walk on water and causes water droplets to form beads on a surface.

To measure the surface tension of CO₂, several methods can be used, including:

1. **Maximum bubble pressure method:** This method involves measuring the pressure required to force a gas bubble through a liquid. The surface tension is calculated from the maximum pressure recorded.

2. Drop weight method: This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is calculated from the weight of the droplet and the radius of the tube.
3. Wilhelmy plate method: This method involves immersing a thin, flat plate into a liquid and measuring the force required to detach it. The surface tension is calculated from the force and the dimensions of the plate.

These are some of the methods that can be used to measure the surface tension of CO₂. It is important to note that the surface tension of CO₂ can vary depending on factors such as temperature and pressure. Therefore, it is important to control these variables when conducting measurements.

Viscosity, Conductance of Solution, Chloride and Iron Content in Water

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- The viscosity of formation water is a function of pressure, temperature, and dissolved solids.
- In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature.
- Dissolved gas in the formation water at reservoir conditions generally results in a negligible effect on water viscosity .

Conductance of Solution

- Conductance is the ability of a solution to conduct an electric current.
- The more ions present in the water, the higher the conductivity.
- High-quality deionized water has a conductivity of about 0.05 S/cm at 25 °C.
- Typical drinking water is in the range of 200–800 S/cm.
- Sea water is about 50 mS/cm (or 0.05 S/cm) .

Chloride Content in Water

- Chloride ions are approximately 1.9 percent of the mass of seawater.
- The more chloride ions present in the water, the higher the conductivity.
- In general, the conductivity of waterways in the United States varies from 50 to 1500 µmhos/cm.
- Inland freshwater lake studies reveal a conductivity of 150 to 500 µmhos/cm .

Iron Content in Water

- Iron is a naturally occurring element in water.
- Iron can be present in water in two forms: soluble ferrous iron or insoluble ferric iron.
- Water containing ferrous iron is clear and colorless because the iron is completely dissolved.
- When exposed to air in the pressure tank or atmosphere, the water turns cloudy and a reddish-brown substance begins to form.
- This sediment is the oxidized or ferric form of iron that will not dissolve in water.

K3

K3 can refer to several things, including a synthetic form of vitamin K and a tax form. Here is some information on both:

1. **Vitamin K3:** Vitamin K3, also known as menadione, is a synthetic or artificially produced form of vitamin K. This is unlike the other two forms of vitamin K — vitamin K1, known as phyloquinone, and vitamin K2, called menaquinone, which occur naturally. Vitamin K3 may be converted into K2 in your liver. Vitamin K is important for blood clotting and bone health.
2. **Schedule K-3 (Form 1065):** Schedule K-3 (Form 1065) is a tax form used by the International Department of the Treasury Internal Revenue Service to report a partner's share of income, deductions, credits, etc.

CO-3 Measure the hardness and alkalinity of the water

Hardness and alkalinity are two important parameters that are used to measure the quality of water. Here are some key points to understand about these two measurements:

1. **Hardness** refers to the concentration of calcium and magnesium ions in water. These ions can come from dissolved minerals such as limestone and dolomite.
2. **Alkalinity** refers to the ability of water to neutralize acids. It is a measure of the buffering capacity of water, which is determined by the presence of bicarbonate, carbonate, and hydroxide ions.
3. To measure the hardness of water, a titration method can be used. This involves adding a chemical reagent to a water sample until a color change occurs, indicating that all the calcium and magnesium ions have reacted.

4. To measure the alkalinity of water, a similar titration method can be used. In this case, a chemical reagent is added to the water sample until a pH change occurs, indicating that all the bicarbonate, carbonate, and hydroxide ions have reacted.
5. The results of these tests are usually expressed in milligrams per liter (mg/L) or parts per million (ppm). Hardness is typically reported as mg/L of calcium carbonate (CaCO_3), while alkalinity is reported as mg/L of bicarbonate (HCO_3^-).
6. The hardness and alkalinity of water can affect its taste, as well as its ability to form scale or corrode pipes and fixtures. Water with high hardness can have a metallic taste, while water with high alkalinity can have a bitter taste.
7. It is important to regularly measure the hardness and alkalinity of water to ensure that it is safe and suitable for its intended use. This can be done using test kits or by sending a water sample to a laboratory for analysis.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that is also known as carbolic acid. It is a weak acid that mostly forms phenoxide ions by dropping one positive hydrogen ion (H^+) from the hydroxyl group. There are several methods for the preparation of phenol, including:

1. **From Haloarenes:** Chlorobenzene is an example of a haloarene which is formed by the monosubstitution of the benzene ring. When chlorobenzene is fused with sodium hydroxide at 623K and 320 atm, sodium phenoxide is produced .
2. **From Benzene Sulphonic Acid:** At 573 – 623K, sodium phenoxide is produced by fusing the sodium salt of benzene sulphonic acid with sodium hydroxide. Sodium phenoxide is converted to phenol when it is treated with dilute acid or carbon dioxide.
3. **From Coal Tar:** Phenol is extracted from the middle oil fraction for commercial purposes at 443 – 503K.
4. **Raschig's Process:** The reaction of benzene, hydrogen chloride and air at 523K produces phenol.
5. **From Cumene:** Phenol can be prepared in large quantities by the air oxidation of cumene.
6. **From Diazonium Salts:** Phenol can be prepared in small amounts by the hydrolysis of diazonium salts.

7. **From Phenylboronic Acid:** Phenol can be prepared in small amounts by the peroxide oxidation of phenylboronic acid.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

Formaldehyde

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.

Urea Formaldehyde Resin

- Urea-formaldehyde (UF) is a nontransparent thermosetting resin or polymer produced from urea and formaldehyde.
- It is synthesized by the “alkali-acid-alkali” method .
- Unlike most thermosetting resins, UF often shows the appearance of crystal domains .
- UF resins are used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.

Adipic Acid

- Adipic acid dihydrazide can be used as a modifier to reduce the free formaldehyde in UF resin.

Paracetamol

- Paracetamol is not mentioned in the search results.

K3

- K3 is a term that can refer to several different things, depending on the context in which it is used.
- One common use of the term K3 is to refer to Vitamin K3, also known as Menadione. It is a synthetic, artificially produced form of vitamin K that doesn't occur naturally.
- Unlike the other two forms of vitamin K — vitamin K1, known as phylloquinone, and vitamin K2, called menaquinone, Vitamin K3 may be converted into K2 in your liver.
- Vitamin K is important for blood clotting and bone health.
- Another use of the term K3 is to refer to Schedule K-3 (Form 1065) of the IRS tax forms. It is used to report a partner's share of income, deductions, credits, etc. for international tax purposes.

CO-5 Estimate the rate constant of reaction. K3

- The rate constant of a reaction, denoted by k , is a proportionality constant that relates the rate of a chemical reaction to the concentrations of the reactants.
- For a termolecular reaction, the reaction rate is described by $r = k_3[A][B][C]$, where k_3 is a termolecular rate constant.
- The rate constant can be calculated by rearranging the terms in the rate equation. For example, for a reaction with the rate equation $\text{Rate} = k[A]^a[B]^b$, the rate constant is calculated as $k = \text{Rate} / ([A]^a[B]^b)$.
- The equilibrium constant for a reaction that is the sum of two or more reactions is equal to the product of the equilibrium constants for the individual reactions.

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- I'm sorry, but I need more information to provide you with the content you are looking for.
- Could you please specify the subject or topic that you want me to write about?

ENGINEERING PHYSICS LAB

Engineering Physics Lab is a course that is designed to provide students with hands-on experience in conducting experiments and making observations in the field of physics. The course is typically taken by students who are pursuing a degree in engineering or a related field.

Some of the key objectives of the Engineering Physics Lab course include:

1. To provide students with an understanding of the basic principles of physics and their applications in engineering.
2. To develop students' skills in conducting experiments, making observations, and analyzing data.
3. To enhance students' ability to work in a team and to communicate their findings effectively.
4. To provide students with an opportunity to apply their knowledge of physics to real-world problems.

The course typically includes a series of experiments that cover topics such as mechanics, thermodynamics, electricity and magnetism, and optics. Students are expected to work in teams to conduct the experiments, collect data, and analyze their results. They are also required to write lab reports and give presentations on their findings.

Overall, the Engineering Physics Lab course is an essential component of an engineering education, as it provides students with the practical skills and knowledge

that they need to succeed in their future careers.

Course Objectives:

1. To provide students with a comprehensive understanding of the subject matter.
2. To develop critical thinking and analytical skills.
3. To enhance the ability to communicate effectively, both orally and in writing.
4. To foster an appreciation for the subject and its relevance to the real world.
5. To prepare students for further study or professional development in the field.
6. To encourage independent learning and self-motivation.
7. To promote collaboration and teamwork.
8. To provide opportunities for practical application of knowledge and skills.
9. To instill ethical and professional values.
10. To assess student learning and provide feedback for improvement.

Analytical Instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are used in various fields such as chemistry, biology, physics, and engineering to provide accurate and reliable data.

1. Fundamental concepts of analytical instruments:
 - **Accuracy:** The degree to which the result of a measurement, calculation, or specification conforms to the correct value or a standard.
 - **Precision:** The degree to which repeated measurements under unchanged conditions show the same results.
 - **Sensitivity:** The capability of an instrument to respond to small changes in the quantity being measured.
 - **Selectivity:** The ability of an instrument to distinguish between different components in a sample.
 - **Resolution:** The smallest difference between two values that can be distinguished by an instrument.
 - **Calibration:** The process of establishing the relationship between the output of an instrument and the value of the input quantity.
 - **Linearity:** The ability of an instrument to produce a linear response over a range of input values.
 - **Reproducibility:** The degree to which repeated measurements of the same sample under the same conditions produce the same results.
 - **Robustness:** The ability of an instrument to maintain its performance under varying conditions.

These concepts are essential for understanding the performance and limitations of analytical instruments and for ensuring that the data obtained from these instruments is accurate and reliable. It is important for students to have a solid understanding of these concepts in order to effectively use and interpret the results obtained from analytical instruments.

Analysis of Chloride Content, Hardness, and Alkalinity of Water

1. **Chloride Content:** Chloride content in water is an important parameter to measure as it can indicate the presence of contamination from human or animal waste, industrial pollution, or saltwater intrusion. The chloride content of water can be measured using various methods, including titration with silver nitrate, ion chromatography, and ion-selective electrodes.
2. **Hardness:** Hardness in water is caused by the presence of dissolved minerals, primarily calcium and magnesium. Hard water can cause scaling in pipes and appliances, and can also interfere with the effectiveness of soaps and detergents. The hardness of water can be measured using titration with ethylenediaminetetraacetic acid (EDTA) or by using a colorimetric test kit.
3. **Alkalinity:** Alkalinity is a measure of the water's ability to neutralize acids. It is primarily caused by the presence of bicarbonate, carbonate, and hydroxide ions. Alkalinity is important to measure as it can affect the pH of the water and the effectiveness of water treatment processes. The alkalinity of water can be measured using titration with a strong acid, such as sulfuric acid or hydrochloric acid.

By understanding the analysis of these parameters, students can gain a better understanding of water quality and the factors that affect it. This knowledge can be applied in various fields, including environmental science, water treatment, and public health.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- **pH** is a measure of the acidity or basicity of a solution. It is defined as the negative logarithm of the hydrogen ion concentration in a solution. The pH scale ranges from 0 to 14, with 7 being neutral. A pH less than 7 is acidic, while a pH greater than 7 is basic. The pH of a solution can be measured using a pH meter or by using pH indicators such as litmus paper.

- **Surface tension** is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the liquid molecules at the surface. Surface tension can be measured using various methods such as the drop weight method, the maximum bubble pressure method, and the Wilhelmy plate method.
- **Viscosity** is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate in a fluid. Viscosity can be measured using various methods such as the capillary tube method, the falling ball method, and the rotational viscometer method.

These properties of liquids are important in various fields such as chemistry, physics, and engineering. Understanding these properties can help students in their studies and future careers.

4. To enable the students to understand about the preparation of different resins.

Resins are solid or highly viscous substances that are typically convertible into polymers. They are usually mixtures of organic compounds, mainly terpenes and their derivatives. Here are some points to understand about the preparation of different resins:

1. **Natural resins:** Natural resins are typically obtained from plants. They are exuded from trees as a response to injury and are collected by tapping, cutting, or making incisions in the bark. Examples of natural resins include amber, copal, dammar, and mastic.
2. **Synthetic resins:** Synthetic resins are materials with similar properties to natural resins but are produced through chemical reactions. They are typically derived from simple hydrocarbons, such as ethylene or propylene, and are often used in the production of plastics, adhesives, and coatings.
3. **Preparation methods:** The preparation of resins varies depending on the type of resin being produced. For natural resins, the collection process involves tapping, cutting, or making incisions in the bark of trees. For synthetic resins, the preparation involves chemical reactions, often involving the polymerization of monomers.
4. **Factors affecting resin preparation:** Various factors can affect the preparation of resins, including temperature, pressure, and the presence of catalysts. These factors can influence the rate of chemical reactions and the properties of the resulting resin.
5. **Safety considerations:** The preparation of resins, particularly synthetic resins, can involve the use of hazardous chemicals and high temperatures. It is important to follow safety guidelines and wear appropriate protective equipment when working with these materials.

5. Synthesis of Organic Compounds: Adipic Acid and Paracetamol

Adipic acid and paracetamol are two organic compounds that can be synthesized through both conventional and green routes. Here, we will discuss the methods of synthesis for both compounds.

Adipic Acid

Adipic acid is an organic compound with the formula $(\text{CH}_2)_4(\text{COOH})_2$. It is a white crystalline powder that is used primarily as a precursor to the production of nylon.

Conventional Route

The conventional route for the synthesis of adipic acid involves the oxidation of cyclohexane with nitric acid. This process produces nitrous oxide (N_2O) as a byproduct, which is a potent greenhouse gas.

Green Route

The green route for the synthesis of adipic acid involves the oxidation of cyclohexene with hydrogen peroxide (H_2O_2) in the presence of a tungsten catalyst. This process produces water as the only byproduct, making it a more environmentally friendly option.

Paracetamol

Paracetamol, also known as acetaminophen, is an organic compound with the formula $\text{C}_8\text{H}_9\text{NO}_2$. It is a widely used over-the-counter pain reliever and fever reducer.

Conventional Route

The conventional route for the synthesis of paracetamol involves the nitration of phenol with nitric acid to produce para-nitrophenol. This is then reduced to para-aminophenol, which is acetylated with acetic anhydride to produce paracetamol.

Green Route

The green route for the synthesis of paracetamol involves the direct acetylation of para-aminophenol with acetic acid in the presence of a solid acid catalyst. This process produces water as the only byproduct, making it a more environmentally friendly option.

In conclusion, both adipic acid and paracetamol can be synthesized through conventional and green routes. The green routes offer more environmentally friendly options, producing water as the only byproduct. It is important for

students to understand these methods of synthesis and the benefits of using green routes.

LIST OF EXPERIMENTS

1. **Milgram Experiment:** A study on obedience to authority figures, conducted by Stanley Milgram in the 1960s.
2. **Stanford Prison Experiment:** A study of the psychological effects of becoming a prisoner or prison guard, conducted by Philip Zimbardo in 1971.
3. **Pavlov's Dogs:** A study on classical conditioning, conducted by Ivan Pavlov in the 1890s.
4. **Asch Conformity Experiment:** A study on conformity, conducted by Solomon Asch in the 1950s.
5. **Hawthorne Effect:** A study on the effect of changes in working conditions on productivity, conducted by Elton Mayo in the 1920s.
6. **Bobo Doll Experiment:** A study on the effects of observed aggression, conducted by Albert Bandura in the 1960s.
7. **Little Albert Experiment:** A study on classical conditioning of fear, conducted by John B. Watson and Rosalie Rayner in 1920.
8. **The Marshmallow Test:** A study on delayed gratification, conducted by Walter Mischel in the 1960s.
9. **The Invisible Gorilla:** A study on inattention blindness, conducted by Christopher Chabris and Daniel Simons in 1999.
10. **The Trolley Problem:** A thought experiment in ethics, first introduced by Philippa Foot in 1967.

1. Calibration of Analytical Equipment and Apparatus

Calibration is the process of verifying the accuracy of a measuring instrument by comparing it with a reference standard. This is important for ensuring that the results obtained from the instrument are accurate and reliable.

- Calibration is necessary for all types of analytical equipment and apparatus, including balances, pipettes, thermometers, and spectrophotometers.
- The frequency of calibration depends on the instrument and its usage. Some instruments may require daily calibration, while others may only need to be calibrated once a year.
- Calibration can be performed by the user, a trained technician, or an external calibration service.
- Calibration involves comparing the readings of the instrument with those of a reference standard. Any discrepancies between the two are corrected by adjusting the instrument.
- Calibration records should be maintained to demonstrate that the instrument has been properly calibrated and is functioning correctly.
- Regular calibration is essential for ensuring the accuracy and reliability of analytical results. It helps to identify any problems with the instrument

and can prevent errors in the analysis.

2. Determination of Hardness of water sample by EDTA method

- Hardness of water is a measure of the concentration of calcium and magnesium ions present in the water.
- The EDTA (Ethylenediaminetetraacetic acid) method is a common method used to determine the hardness of water.
- In this method, a solution of EDTA is used as a titrant to react with the calcium and magnesium ions present in the water sample.
- The end point of the titration is determined using an indicator such as Eriochrome Black T, which changes color when all the calcium and magnesium ions have reacted with the EDTA.
- The volume of EDTA solution used in the titration is then used to calculate the hardness of the water sample.
- This method is simple, accurate, and widely used in the analysis of water samples.

3. Determination of Alkalinity of water sample

Alkalinity is a measure of the ability of water to neutralize acids. It is an important parameter in water quality analysis, as it provides information about the buffering capacity of water. Here are the steps to determine the alkalinity of a water sample:

1. Collect a water sample in a clean container.
2. Measure the pH of the sample using a pH meter or pH test strips.
3. If the pH is above 8.3, the sample is considered to have high alkalinity.
4. If the pH is between 4.5 and 8.3, titrate the sample with a standard acid solution, such as sulfuric acid or hydrochloric acid, using a burette.
5. Record the volume of acid required to lower the pH to 4.5.
6. Calculate the alkalinity of the sample using the formula: $\text{Alkalinity (mg/L as CaCO}_3\text{)} = (\text{Volume of acid used} * \text{Normality of acid} * 50,000) / \text{Volume of sample}$.

This is a basic overview of the process of determining the alkalinity of a water sample. It is important to follow standard procedures and use calibrated equipment to ensure accurate results.

4. Determination of pH by titrimetric method

Titrimetric method is a common laboratory technique used to determine the pH of a solution. The process involves the following steps:

1. A known volume of the solution whose pH is to be determined is taken in a conical flask.
2. A suitable indicator is added to the solution. The choice of indicator depends on the pH range of the solution.

3. A standard solution of a strong acid or a strong base is taken in a burette. The choice of acid or base depends on whether the solution is acidic or basic.
4. The standard solution is slowly added to the solution in the conical flask with constant stirring.
5. The addition of the standard solution is continued until the indicator changes color, indicating the end point of the titration.
6. The volume of the standard solution used is noted.
7. The pH of the solution can be calculated using the formula $\text{pH} = -\log[\text{H}^+]$, where $[\text{H}^+]$ is the concentration of hydrogen ions in the solution.

This method is simple, accurate, and can be used to determine the pH of a wide range of solutions. However, it requires the use of a suitable indicator and a standard solution of known concentration. Additionally, the end point of the titration must be accurately determined to obtain accurate results.

5. Determination of surface tension of given liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the molecules at the surface of a liquid. It is the force that acts along the surface of the liquid and tends to minimize the surface area. There are several methods to determine the surface tension of a given liquid, including:

1. **Capillary rise method:** This method involves measuring the height to which a liquid rises in a narrow tube due to the surface tension of the liquid. The surface tension is then calculated using the formula: $\text{Surface tension} = (\text{weight of liquid raised} \times \text{radius of the tube}) / \text{height of liquid rise}$.
2. **Maximum bubble pressure method:** This method involves measuring the maximum pressure required to blow a bubble from the end of a submerged tube. The surface tension is then calculated using the formula: $\text{Surface tension} = (4 \times \text{maximum pressure} \times \text{radius of the tube}) / 3$.
3. **Drop weight method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is then calculated using the formula: $\text{Surface tension} = (\text{weight of droplet} \times \text{acceleration due to gravity}) / (2 \times \pi \times \text{radius of the tube})$.
4. **Wilhelmy plate method:** This method involves immersing a thin flat plate into a liquid and measuring the force required to detach it from the liquid. The surface tension is then calculated using the formula: $\text{Surface tension} = \text{force} / \text{perimeter of the plate}$.
5. **Du Nouy ring method:** This method involves immersing a thin ring into a liquid and measuring the force required to detach it from the liquid. The surface tension is then calculated using the formula: $\text{Surface tension} = (\text{force} - \text{weight of the ring}) / (2 \times \pi \times \text{radius of the ring})$.

tension = (force - weight of ring in air + weight of ring in liquid) / circumference of the ring.

Each of these methods has its own advantages and limitations, and the choice of method depends on the specific requirements of the experiment. It is important to carefully calibrate the instruments and follow the experimental procedure to obtain accurate results.

6. Determination of Viscosity of a given liquid by viscometer

Viscosity is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate. A viscometer is an instrument used to measure the viscosity of a fluid. There are several types of viscometers, including capillary, rotational, and falling ball viscometers.

To determine the viscosity of a given liquid using a viscometer, the following steps can be followed:

1. **Calibration:** The viscometer should be calibrated using a liquid of known viscosity, such as water, to ensure accurate measurements.
2. **Sample preparation:** The liquid to be tested should be prepared according to the instructions of the viscometer. This may include filtering, degassing, or heating the liquid to a specific temperature.
3. **Measurement:** The liquid is placed in the viscometer and the viscosity is measured according to the instructions of the instrument. This may involve measuring the time it takes for a ball to fall through the liquid, the torque required to rotate a spindle in the liquid, or the pressure drop across a capillary tube.
4. **Calculation:** The viscosity of the liquid can be calculated using the measurements obtained and the calibration data.

It is important to follow the instructions of the viscometer carefully to ensure accurate measurements. Factors such as temperature, sample preparation, and instrument calibration can all affect the accuracy of the viscosity measurement.

7. Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. It is commonly used as a primary standard in redox titrations due to its high purity and stability.
3. The strength of a solution of ferrous ammonium sulfate can be determined by titrating it against a standard solution of potassium permanganate (KMnO_4) using an external indicator.
4. The external indicator used in this titration is potassium ferricyanide ($\text{K}_3\text{Fe}(\text{CN})_6$).

5. The reaction between ferrous ammonium sulfate and potassium permanganate is a redox reaction, where potassium permanganate is reduced to manganese sulfate (MnSO_4) and ferrous ammonium sulfate is oxidized to ferric ammonium sulfate.
6. The end point of the titration is indicated by the appearance of a blue color, due to the formation of Prussian blue, when all the ferrous ions have been oxidized to ferric ions.
7. The strength of the ferrous ammonium sulfate solution can be calculated using the volume of potassium permanganate solution used and the molarity of the potassium permanganate solution.

8. Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is an oxidizing agent that can be used to determine the strength of a substance.
2. An internal indicator is used to determine the endpoint of the titration.
3. The internal indicator changes color when the reaction is complete, indicating that the titration is finished.
4. The strength of the potassium dichromate can then be calculated using the volume of the titrant and the concentration of the substance being titrated.
5. This method is commonly used in analytical chemistry to determine the strength of a substance.

Determination of available chlorine in bleaching powder

Bleaching powder, also known as calcium hypochlorite, is a chemical compound that is commonly used as a disinfectant and bleaching agent. The amount of available chlorine in bleaching powder is an important parameter that determines its effectiveness as a disinfectant.

Here are the steps to determine the available chlorine in bleaching powder:

1. Weigh a sample of bleaching powder and dissolve it in water to make a solution.
2. Take a known volume of the solution and add an excess of potassium iodide (KI) to it. This will cause the iodine to be liberated.
3. Titrate the liberated iodine with a standard solution of sodium thiosulfate using starch as an indicator.
4. The end point of the titration is indicated by the disappearance of the blue color of the starch-iodine complex.
5. From the volume of sodium thiosulfate used, the amount of iodine can be calculated.

6. The amount of available chlorine in the bleaching powder can then be calculated using the following formula: Available chlorine (g) = (Iodine liberated (g) x 35.5) / 80.

This method is commonly used to determine the available chlorine in bleaching powder and is based on the principle of iodometry. It is a simple and accurate method that can be easily performed in a laboratory setting.

10. Determination of chloride content in water sample

1. Chloride content in water can be determined using a titration method known as the Mohr method.
2. The Mohr method involves titrating the water sample with a silver nitrate solution.
3. The silver nitrate reacts with the chloride ions in the water to form a white precipitate of silver chloride.
4. The end point of the titration is indicated by the formation of a red-brown precipitate of silver chromate.
5. The amount of silver nitrate used in the titration is used to calculate the concentration of chloride ions in the water sample.
6. The Mohr method is a simple and accurate method for determining the chloride content in water samples.
7. It is important to determine the chloride content in water as high levels of chloride can be harmful to aquatic life and can also affect the taste of drinking water.
8. Other methods for determining the chloride content in water include ion chromatography and ion-selective electrodes.
9. These methods are more sensitive and can detect lower levels of chloride in water samples.
10. However, they require more specialized equipment and are more complex to perform than the Mohr method.

11. Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol with formaldehyde. Here are the steps involved in the preparation of PF resin:

1. **Phenol and formaldehyde are mixed:** The first step in the preparation of PF resin is to mix phenol and formaldehyde in the presence of an acidic or basic catalyst. The ratio of formaldehyde to phenol is typically between 1:1 and 3:1.
2. **Formation of hydroxymethylphenols:** The reaction between phenol and formaldehyde produces hydroxymethylphenols, which can further react with each other or with more formaldehyde to form more complex compounds.

3. **Formation of resin:** As the reaction progresses, the hydroxymethylphenols react with each other to form larger molecules, eventually forming a resinous material. The reaction can be controlled by adjusting the temperature, pH, and catalyst concentration.
4. **Curing:** The final step in the preparation of PF resin is curing, which involves heating the resin to a high temperature to complete the polymerization process. This results in a hard, durable material that is resistant to heat and chemicals.

PF resins are widely used in the production of laminates, adhesives, and molded objects due to their excellent mechanical and thermal properties. They are also used as binders in the production of wood-based panels such as plywood and particleboard.

12. Preparation of Urea formaldehyde (UF) resin

Urea formaldehyde (UF) resin is a type of thermosetting polymer that is commonly used in the production of adhesives, finishes, and molded objects. Here are the steps involved in the preparation of UF resin:

1. **Reacting Urea and Formaldehyde:** The first step in the preparation of UF resin is the reaction of urea and formaldehyde. This reaction takes place in an aqueous solution, with the molar ratio of urea to formaldehyde typically ranging from 1:1.5 to 1:2.5. The reaction is carried out at a pH of 7-8 and a temperature of 70-90°C.
2. **Condensation:** The next step is the condensation of the reaction product. This is achieved by heating the reaction mixture to a temperature of 90-100°C and adjusting the pH to 4.5-5.5. This causes the formation of methylene and methylene ether linkages between the urea molecules, resulting in the formation of a viscous resin.
3. **Concentration:** The resin is then concentrated by removing excess water through evaporation. This is typically done under vacuum at a temperature of 60-80°C.
4. **Curing:** The final step in the preparation of UF resin is curing. This involves the addition of a curing agent, such as ammonium chloride or ammonium sulfate, to the resin. The curing agent causes the resin to crosslink, forming a hard, insoluble polymer. The curing process typically takes place at a temperature of 100-130°C.

These are the basic steps involved in the preparation of UF resin. The exact conditions and proportions may vary depending on the specific application and desired properties of the final product.

Preparation of Adipic acid / Paracetamol

Adipic acid and paracetamol are two different compounds with different preparation methods.

Adipic acid

Adipic acid is an organic compound with the formula $(\text{CH}_2)_4(\text{COOH})_2$. It is a white crystalline powder that is mainly used as a precursor to the production of nylon.

1. Adipic acid can be prepared by the oxidation of cyclohexanol or cyclohexanone with nitric acid.
2. Another method for the preparation of adipic acid is the oxidation of cyclohexene with a mixture of potassium permanganate and sulfuric acid.
3. Adipic acid can also be prepared by the hydrolysis of acrylonitrile followed by the oxidation of the resulting adiponitrile with nitric acid.

Paracetamol

Paracetamol, also known as acetaminophen, is a medication used to treat pain and fever. It is typically used for mild to moderate pain relief.

1. Paracetamol can be prepared by the nitration of phenol followed by the reduction of the resulting 4-nitrophenol with sodium borohydride.
2. Another method for the preparation of paracetamol is the acetylation of 4-aminophenol with acetic anhydride.
3. Paracetamol can also be prepared by the hydrolysis of p-acetanide followed by the acetylation of the resulting p-aminophenol with acetic anhydride.

14. Determination of Cell Conductance of a solution

1. Cell conductance is a measure of a solution's ability to conduct electricity.
2. It is determined by measuring the electrical conductivity of the solution using a conductivity meter.
3. The conductivity meter consists of a cell with two electrodes and a circuit to measure the current passing through the solution.
4. The cell constant is determined by measuring the conductivity of a solution of known concentration.
5. The cell conductance is then calculated by dividing the measured conductivity by the cell constant.
6. The cell conductance is affected by the concentration, temperature, and nature of the ions present in the solution.
7. It is important to calibrate the conductivity meter and to control the temperature of the solution during the measurement to obtain accurate results.

15. Determination of Rate constant of hydrolysis of esters

1. The hydrolysis reaction of an ester in pure water is a slow reaction .
2. When a mineral acid like hydrochloric acid is added, the rate of the reaction is enhanced since the H^+ ions from the mineral acid acts as the catalyst .
3. The hydrolysis of esters is catalyzed by either an acid or a base .
4. Acidic hydrolysis is simply the reverse of esterification .
5. The ester is heated with a large excess of water containing a strong-acid catalyst .
6. Like esterification, the reaction is reversible and does not go to completion .
7. The rate constant for the hydrolysis reaction of an ester by dilute acid is $0.6931 \times 10^{-3} \text{ s}^{-1}$.
8. The rate constant after calculation from the graphs was approximately $0.003 \text{ min}^{-1} \text{ cm}^{-3}$ for the 1ml and 2ml of ethyl acetate .

16. Element detection and identification of functional groups in organic compounds

Organic compounds are composed of carbon, hydrogen, and other elements such as nitrogen, oxygen, sulfur, and halogens. The presence of these elements can be detected through various tests.

1. **Detection of Carbon and Hydrogen:** Carbon and hydrogen are detected by heating the organic compound with copper oxide in a test tube. Carbon is oxidized to carbon dioxide, which turns lime water milky, while hydrogen is oxidized to water, which can be detected by the presence of moisture on the cooler parts of the test tube.
2. **Detection of Nitrogen:** Nitrogen can be detected by the Lassaigne's test. In this test, the organic compound is fused with sodium metal, which converts nitrogen into sodium cyanide. The sodium cyanide is then extracted with water and reacted with ferrous sulfate and ferric chloride to form Prussian blue, indicating the presence of nitrogen.
3. **Detection of Oxygen:** Oxygen is not directly detected in organic compounds. Instead, its presence is inferred from the absence of other elements.
4. **Detection of Sulfur:** Sulfur can be detected by the sodium fusion test, similar to the Lassaigne's test for nitrogen. The sulfur in the organic compound is converted to sodium sulfide, which is extracted with water and reacted with sodium nitroprusside to form a violet color, indicating the presence of sulfur.
5. **Detection of Halogens:** Halogens can be detected by the Beilstein test. In this test, the organic compound is heated with copper wire in a flame.

The halogens react with copper to form copper halides, which impart a characteristic color to the flame.

Functional groups are specific arrangements of atoms within an organic molecule that confer characteristic chemical properties. Some common functional groups and their identification tests include:

1. **Alcohols:** Alcohols contain the hydroxyl (-OH) functional group. They can be identified by the Lucas test, in which the alcohol is reacted with Lucas reagent (a mixture of zinc chloride and hydrochloric acid) to form a cloudy solution.
2. **Aldehydes and Ketones:** Aldehydes and ketones contain the carbonyl ($>\text{C}=\text{O}$) functional group. They can be identified by the 2,4-dinitrophenylhydrazine (2,4-DNP) test, in which the aldehyde or ketone reacts with 2,4-DNP to form a yellow or orange precipitate.
3. **Carboxylic Acids:** Carboxylic acids contain the carboxyl (-COOH) functional group. They can be identified by their reaction with sodium bicarbonate to produce carbon dioxide gas, which can be detected by its effervescence.
4. **Amines:** Amines contain the amino (-NH₂) functional group. They can be identified by their reaction with nitrous acid to form nitrogen gas, which can be detected by its effervescence.

These are some of the methods used for element detection and identification of functional groups in organic compounds. It is important to note that these tests are not always conclusive and may require further analysis for confirmation.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose any 10 experiments from the list provided.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read and understand the instructions for each experiment.
- Students should also be prepared for the possibility that the instructor may change some of the experiments.
- It is recommended that students communicate with the instructor to clarify any questions or concerns they may have about the experiments.

Course Outcomes:

1. Understanding the fundamental concepts and principles of the subject.
2. Developing the ability to apply the learned concepts and principles to solve problems.
3. Enhancing critical thinking and analytical skills.
4. Improving communication skills through written and oral presentations.
5. Developing the ability to work effectively in a team.
6. Gaining an appreciation for the importance of lifelong learning and professional development.
7. Acquiring the skills and knowledge necessary to pursue further studies or a career in the field.

Upon completion of the course the student should be able to:

1. Demonstrate a thorough understanding of the course material.
2. Apply the concepts and theories learned in the course to real-world situations.
3. Analyze and evaluate information critically and effectively.
4. Communicate ideas and arguments clearly and effectively in both written and oral forms.
5. Work collaboratively with others to achieve common goals.
6. Demonstrate ethical and professional behavior in all course-related activities.
7. Use technology effectively to enhance learning and communication.
8. Engage in self-directed learning and continuous improvement.
9. Demonstrate an appreciation for diversity and multiculturalism.
10. Develop and apply problem-solving skills to complex issues.

Course Outcomes Bloom's

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is commonly used to design curriculum and assess student learning. The taxonomy is divided into six levels, with each level building upon the previous one. The six levels are:

1. **Remembering:** This level involves recalling information from memory. It is the most basic level of learning and includes tasks such as defining, listing, and recognizing.
2. **Understanding:** This level involves comprehending the meaning of information. It includes tasks such as explaining, summarizing, and interpreting.
3. **Applying:** This level involves using information in a new situation. It includes tasks such as solving problems, demonstrating, and implementing.
4. **Analyzing:** This level involves breaking down information into parts and understanding their relationships. It includes tasks such as comparing,

contrasting, and categorizing.

5. **Evaluating:** This level involves making judgments about the value of information. It includes tasks such as critiquing, justifying, and assessing.
6. **Creating:** This level involves combining information to form a new whole. It includes tasks such as designing, constructing, and planning.

Course outcomes are statements that describe what students should know or be able to do by the end of a course. These outcomes are often aligned with Bloom's Taxonomy to ensure that students are achieving higher levels of learning. For example, a course outcome might state that by the end of the course, students will be able to analyze and evaluate arguments in a particular field of study. This outcome aligns with the analyzing and evaluating levels of Bloom's Taxonomy. By designing course outcomes that align with Bloom's Taxonomy, educators can ensure that their courses are promoting higher levels of learning and critical thinking.

Level

- Level is a term used to describe a relative position or degree of something.
- In mathematics, level can refer to a specific value on a scale or a specific height on a graph.
- In geography, level can refer to the height of a location relative to sea level.
- In gaming, level can refer to a stage or difficulty of a game.
- In education, level can refer to the degree of proficiency or knowledge in a subject.
- In construction, a level is a tool used to determine if a surface is horizontal or vertical.
- In physics, level can refer to the energy state of an atom or molecule.
- In music, level can refer to the volume or intensity of a sound.
- In sports, level can refer to the skill or ability of a player or team.
- In general, level can be used to describe the relative importance or rank of something.

CO-1 Get an understanding of the use of different analytical instruments. K3

Analytical instruments are used to measure physical, chemical, and biological properties of substances. These instruments are used in a variety of fields, including chemistry, biology, physics, and engineering. Some common analytical instruments include:

1. **Spectrophotometers:** These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.

2. **Chromatographs:** These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.
3. **Mass spectrometers:** These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.
4. **Electrochemical instruments:** These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.
5. **Microscopes:** These instruments magnify small objects to make them visible to the naked eye. They are commonly used to study the structure of cells and tissues.

Each of these instruments has its own unique capabilities and limitations. Understanding the use of different analytical instruments is essential for selecting the appropriate instrument for a given task and for interpreting the results obtained from the instrument.

CO-2 Measure the molecular / system properties such as surface tension

Surface tension is a property of liquids that arises due to the cohesive forces between the molecules at the surface of the liquid. These forces cause the liquid to behave as if it is covered by an elastic film, which resists deformation and tends to minimize the surface area.

Some methods to measure surface tension include:

1. **Maximum Bubble Pressure Method:** This method involves measuring the pressure required to force a gas bubble out of a submerged capillary tube. The pressure difference between the inside and outside of the bubble is related to the surface tension of the liquid.
2. **Drop Weight Method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The weight of the droplet is related to the surface tension of the liquid.
3. **Wilhelmy Plate Method:** This method involves immersing a thin, flat plate into a liquid and measuring the force required to detach it from the liquid surface. The force is related to the surface tension of the liquid.
4. **Du Nouy Ring Method:** This method involves immersing a ring into a liquid and measuring the force required to detach it from the liquid surface. The force is related to the surface tension of the liquid.

These are some of the common methods used to measure the surface tension of liquids. Each method has its own advantages and limitations, and the choice of method depends on the specific requirements of the experiment.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- The viscosity of formation water is a function of pressure, temperature, and dissolved solids.
- In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature.
- Dissolved gas in the formation water at reservoir conditions generally results in a negligible effect on water viscosity .

Conductance of Solution

- Conductivity is a measure of a solution's ability to conduct an electric current.
- The more chloride ions present in the water, the higher the conductivity.
- In general, the conductivity of waterways in the United States varies from 50 to 1500 $\mu\text{mhos/cm}$, and inland freshwater lake studies reveal a conductivity of 150 to 500 $\mu\text{mhos/cm}$.
- High-quality deionized water has a conductivity of about 0.05 S/cm at 25 °C, typical drinking water is in the range of 200–800 S/cm, while seawater is about 50 mS/cm (or 0.05 S/cm) .

Chloride Content in the Water

- Chloride ions are approximately 1.9 percent of the mass of seawater .

Iron Content in the Water

K3

- K3 is a term that can refer to several different things.
- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a type of visa for spouses of U.S. citizens.
- K3 is also a term used in mathematics to refer to a type of surface.
- In the context of education, K3 may refer to a grade level in a school system.
- Without more context, it is difficult to determine the specific meaning of K3 in this case.

CO-3 Measure the hardness and alkalinity of the water. K3

Hardness and alkalinity are two important parameters of water quality. Hardness refers to the concentration of calcium and magnesium ions in water, while

alkalinity refers to the ability of water to neutralize acids.

1. **Measuring Hardness:** Hardness can be measured using a titration method, where a solution of known concentration is added to a water sample until a color change occurs. The amount of solution required to reach the color change indicates the hardness of the water. Another method is to use a hardness test strip, which changes color based on the hardness of the water.
2. **Measuring Alkalinity:** Alkalinity can also be measured using a titration method, where a solution of known concentration is added to a water sample until a color change occurs. The amount of solution required to reach the color change indicates the alkalinity of the water. Another method is to use an alkalinity test kit, which measures the pH of the water and uses that information to calculate the alkalinity.

It is important to measure the hardness and alkalinity of water to ensure that it is suitable for its intended use. For example, water with high hardness can cause scaling in pipes and appliances, while water with low alkalinity can be corrosive. By measuring these parameters, appropriate treatment can be applied to improve the water quality.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an aromatic organic compound with the molecular formula C_6H_5OH . It is also known as carbolic acid, hydroxybenzene, or phenic acid. Phenol is a white crystalline solid at room temperature and is soluble in water, alcohol, and ether. It has a characteristic odor and is used in the production of many chemicals, including plastics, resins, and pharmaceuticals.

There are several methods for the preparation of phenol, including:

1. **From Cumene:** Cumene is oxidized in the presence of air to produce cumene hydroperoxide, which is then cleaved in the presence of an acid catalyst to produce phenol and acetone.
2. **From Chlorobenzene:** Chlorobenzene is heated with an aqueous solution of sodium hydroxide at high temperature and pressure to produce phenol and sodium chloride.
3. **From Benzene Sulfonic Acid:** Benzene sulfonic acid is fused with an alkali, such as sodium hydroxide, to produce phenol and sodium sulfate.
4. **From Aniline:** Aniline is diazotized in the presence of nitrous acid to produce benzenediazonium chloride, which is then heated with water to produce phenol and nitrogen gas.

These are some of the fundamental concepts of the preparation of phenol. It is important to understand these methods and the underlying chemical reactions in order to have a thorough understanding of the subject.

Formaldehyde, Urea Formaldehyde Resin, Adipic Acid, and Paracetamol

Formaldehyde

Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products. It is used in pressed-wood products, such as particleboard, plywood, and fiberboard; glues and adhesives; permanent-press fabrics; paper product coatings; and certain insulation materials.

Urea Formaldehyde Resin

Urea-formaldehyde (UF), also known as urea-methanal, is a nontransparent thermosetting resin or polymer. It is produced from urea and formaldehyde. These resins are used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects. A stable urea-formaldehyde resin (UF) is synthesized by the “alkali-acid-alkali” method using formaldehyde and urea as raw materials. Unlike most thermosetting resins, UF often shows the appearance of crystal domains .

Adipic Acid

Melamine, polyvinyl alcohol, and adipic acid dihydrazide as modifiers were added to urea-formaldehyde (UF) resin for reducing the free formaldehyde. The influence of addition amount and addition stage of modifiers on the physicochemical property and free formaldehyde content of UF resin was tested.

Paracetamol

K3

K3 is a term that can refer to several different things. Without more context, it is difficult to determine which specific topic you are referring to. Some possible interpretations of K3 include:

1. K3, a Belgian-Dutch girl group formed in 1998.
2. K3, a mountain in the Karakoram range, also known as Broad Peak.
3. K3, a type of visa for spouses of U.S. citizens to enter the United States while waiting for their immigrant visa to be processed.
4. K3, a type of South Korean high-speed train.
5. K3, a type of tax form used in the United States for reporting income from partnerships, S corporations, and trusts.

CO-5 Estimate the rate constant of reaction. K3

- The rate constant of a reaction, denoted by k , is a measure of the speed of a chemical reaction.
- It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers.
- The rate constant is specific to a particular reaction and is dependent on factors such as temperature, pressure, and the presence of a catalyst.
- The rate constant can be determined experimentally by measuring the rate of the reaction at different concentrations of the reactants and then using the rate law to calculate the value of k .
- The units of the rate constant depend on the order of the reaction. For a first-order reaction, the units are s^{-1} , for a second-order reaction, the units are $M^{-1} s^{-1}$, and for a third-order reaction, the units are $M^{-2} s^{-1}$.
- The Arrhenius equation can be used to calculate the rate constant at different temperatures. The equation is $k = Ae^{(-E_a/RT)}$, where A is the pre-exponential factor, E_a is the activation energy, R is the gas constant, and T is the temperature in kelvins.
- The value of the rate constant can also be affected by the presence of a catalyst, which lowers the activation energy of the reaction and increases the rate constant.

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- I'm sorry, but I need more information to provide a response on the topic "Page 31 of 40".
- Can you please specify the context or the subject matter of the page you are referring to?
- This will help me provide a more accurate and detailed response. Thank you.

Engineering Mathematics Lab KCS

Engineering Mathematics Lab KCS is a laboratory course that is designed to provide students with hands-on experience in applying mathematical concepts and techniques to solve engineering problems. The course typically covers topics such as:

1. Numerical methods for solving equations and systems of equations.
2. Techniques for curve fitting and interpolation.
3. Numerical integration and differentiation.
4. Optimization techniques.
5. Probability and statistics.

The course is usually structured around a series of laboratory exercises that allow students to apply the mathematical concepts and techniques they have learned to solve practical engineering problems. The exercises are typically designed to be completed using mathematical software packages such as MATLAB or Mathematica.

The goal of the Engineering Mathematics Lab KCS is to provide students with a deeper understanding of the mathematical concepts and techniques that are used in engineering, and to develop their ability to apply these concepts and techniques to solve practical problems. By completing the laboratory exercises, students gain experience in using mathematical software packages and in working with real-world data, which can be valuable skills for their future careers.

Course Objectives:

- To provide a comprehensive understanding of the subject matter.
- To develop critical thinking and analytical skills.
- To enhance problem-solving abilities.
- To encourage independent learning and research.
- To foster effective communication and collaboration.
- To prepare students for further academic or professional pursuits.
- To promote the application of knowledge in real-world situations.
- To instill a sense of ethical responsibility and social awareness.

Analytical Instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of a sample. These instruments are commonly used in scientific research, quality control, and medical diagnosis.

1. Fundamental Concepts of Analytical Instruments:

- Analytical instruments are designed to provide accurate and precise measurements of a sample's properties.
- These instruments use various techniques and methods to analyze the sample, including spectroscopy, chromatography, and mass spectrometry.
- The choice of analytical instrument and technique depends on the type of sample being analyzed and the information required.
- Proper calibration and maintenance of analytical instruments are essential to ensure accurate and reliable results.
- Data analysis and interpretation are critical components of the analytical process, and require a thorough understanding of the instrument and the underlying principles of the technique used.

Analysis of Chloride Content, Hardness, and Alkalinity

1. **Chloride Content:** Chloride content refers to the amount of chloride ions present in a sample of water. Chloride ions are negatively charged particles that can be found in many types of salts, including sodium chloride (common table salt). The presence of chloride ions in water can indicate contamination from sources such as seawater intrusion, industrial waste, or road salt runoff.
2. **Hardness:** Water hardness is a measure of the concentration of calcium and magnesium ions in water. These ions can cause problems in industrial and domestic settings by forming deposits on pipes and fixtures, reducing the effectiveness of soaps and detergents, and interfering with certain chemical reactions. Water hardness is typically measured in grains per gallon (gpg) or milligrams per liter (mg/L) as calcium carbonate.
3. **Alkalinity:** Alkalinity refers to the ability of water to neutralize acids. It is a measure of the buffering capacity of water, or its ability to resist changes in pH. Alkalinity is important in maintaining a stable pH in aquatic ecosystems and in preventing corrosion in water distribution systems. Alkalinity is typically measured in milligrams per liter (mg/L) as calcium carbonate.

In summary, the analysis of chloride content, hardness, and alkalinity are important in understanding the quality and characteristics of water. These parameters can provide information on potential sources of contamination, the suitability of water for various uses, and the need for treatment to improve water quality.

Water

- Water is a substance composed of the chemical elements hydrogen and oxygen and existing in gaseous, liquid, and solid states.
- It is one of the most plentiful and essential of compounds.
- A tasteless and odourless liquid at room temperature, it has the important ability to dissolve many other substances.
- Water is a tiny molecule. It consists of three atoms: two of hydrogen and one of oxygen.
- Water molecules cling to each other because of a force called hydrogen bonding.
- Water is a shape-shifter. It exists in three states on Earth: liquid, gas, and solid.
- Water is the liquid that makes life on Earth possible.
- As water cycles from the air to the land to the sea and back again, water shapes our planet and nearly every aspect of our lives.
- All living things, from tiny cyanobacteria to giant blue whales, need water

to survive.

- Water is an inorganic compound with the chemical formula H_2O .
- It is a transparent, tasteless, odorless, and nearly colorless chemical substance.
- It is the main constituent of Earth's hydrosphere and the fluids of all known living organisms (in which it acts as a solvent).

3. Measure of pH, Surface Tension and Viscosity

pH

- pH is a measure of the acidity or basicity of a solution.
- It is defined as the negative logarithm of the hydrogen ion concentration in a solution.
- The pH scale ranges from 0 to 14, with 7 being neutral.
- A pH value less than 7 indicates an acidic solution, while a pH value greater than 7 indicates a basic solution.

Surface Tension

- Surface tension is the property of a liquid that allows it to resist an external force.
- It is caused by the cohesive forces between the molecules of the liquid.
- Surface tension is measured in units of force per unit length, typically in millinewtons per meter (mN/m).
- The surface tension of a liquid can be measured using various methods, such as the drop weight method or the maximum bubble pressure method.

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- It is defined as the ratio of the shear stress to the shear rate in a fluid.
- Viscosity is typically measured in units of pascal-seconds ($\text{Pa} \cdot \text{s}$).
- The viscosity of a fluid can be measured using various methods, such as the capillary tube method or the rotating cylinder method.

Liquid

- A liquid is one of the four fundamental states of matter, along with solid, gas, and plasma.
- It is characterized by its ability to conform to the shape of its container while retaining a constant volume.
- Liquids have a definite volume, but no definite shape.
- The particles in a liquid are close together and are in constant motion, allowing them to slide past one another.

- This gives liquids the ability to flow and take the shape of their container.
- The attractive forces between the particles in a liquid are strong enough to keep the particles close together, but not strong enough to keep them in a fixed position.
- This results in a liquid having a definite volume, but no definite shape.
- Liquids have a much higher density than gases, but a lower density than solids.
- The density of a liquid is typically about the same as the density of the solid form of the substance.
- Liquids can be compressed, but not as easily as gases.
- The compressibility of a liquid is much lower than that of a gas, but higher than that of a solid.
- Liquids can also be heated and cooled, which can cause them to change state.
- When a liquid is heated, its particles gain energy and move faster, causing the liquid to expand.
- If the liquid is heated enough, it can change into a gas.
- When a liquid is cooled, its particles lose energy and move slower, causing the liquid to contract.
- If the liquid is cooled enough, it can change into a solid.
- Liquids can also be mixed with other liquids to form solutions.
- The ability of two liquids to mix depends on the attractive forces between the particles of the two liquids.
- If the attractive forces between the particles of the two liquids are similar, the liquids will mix.
- If the attractive forces between the particles of the two liquids are very different, the liquids will not mix and will form two separate layers.
- Liquids can also dissolve solids and gases, forming solutions.
- The ability of a liquid to dissolve a solid or a gas depends on the attractive forces between the particles of the liquid and the particles of the solid or gas.
- If the attractive forces between the particles of the liquid and the particles of the solid or gas are strong enough, the solid or gas will dissolve in the liquid.
- If the attractive forces between the particles of the liquid and the particles of the solid or gas are not strong enough, the solid or gas will not dissolve in the liquid.

Preparation of Different Resins

Resins are solid or highly viscous substances that are typically convertible into polymers. They are used in a variety of applications, including adhesives, coatings, and plastics. Here are some points to consider when preparing different resins:

1. **Type of resin:** There are several types of resins, including natural resins (such as rosin and shellac) and synthetic resins (such as phenolic and epoxy resins). The type of resin used will depend on the desired properties and application.
2. **Raw materials:** The raw materials used to prepare resins will vary depending on the type of resin being prepared. For example, phenolic resins are typically prepared from phenol and formaldehyde, while epoxy resins are prepared from epichlorohydrin and bisphenol A.
3. **Reaction conditions:** The reaction conditions, such as temperature, pressure, and catalyst, will affect the properties of the resulting resin. For example, increasing the temperature can increase the rate of reaction, while the use of a catalyst can improve the efficiency of the reaction.
4. **Purification:** After the resin has been prepared, it may need to be purified to remove any impurities or unreacted raw materials. This can be done through methods such as filtration, distillation, or extraction.
5. **Formulation:** The final resin may be formulated with other materials, such as fillers, pigments, or additives, to improve its properties or performance in a specific application.

In summary, the preparation of different resins involves selecting the appropriate type of resin, raw materials, and reaction conditions, as well as purifying and formulating the final product. These steps must be carefully controlled to produce a resin with the desired properties and performance.

Synthesis of Adipic Acid

Adipic acid is an important organic compound that is used in the production of nylon, polyurethane, and other industrial products. Here are some key points to understand about the synthesis of adipic acid:

1. Adipic acid is typically synthesized from cyclohexane or cyclohexanone through a process known as oxidation.
2. The oxidation of cyclohexane involves the use of a catalyst, typically cobalt or manganese, and an oxidizing agent, such as nitric acid or oxygen.
3. The oxidation of cyclohexanone, on the other hand, can be achieved through the use of a variety of oxidizing agents, including nitric acid, hydrogen peroxide, and oxygen.
4. The resulting product of the oxidation reaction is a mixture of adipic acid and other byproducts, which must be separated and purified to obtain pure adipic acid.
5. Alternative methods for the synthesis of adipic acid include the carbonylation of butadiene and the hydrocarboxylation of cyclohexene.

These are some of the key points to understand about the synthesis of adipic

acid. It is an important industrial process that enables the production of a variety of useful materials.

Acid and Paracetamol by Conventional and Green Route

1. Paracetamol, also known as N-acetyl-p-aminophenol (PAR), and acetylsalicylic acid (aspirin, ASA) can be synthesized using principles of green chemistry.
2. An alternative route for the synthesis of these compounds involves solvent-free synthesis combined with microwave-energy irradiation .
3. In contrast, the conventional method of synthesizing paracetamol involves the use of solvents and may not be as environmentally friendly.
4. Green chemistry aims to reduce the environmental impact of chemical processes by minimizing the use of hazardous substances and maximizing atom economy.
5. For example, in a green synthesis of acetaminophen from phenol, the solvent from step two is kept to help maximize atom economy.

List of Experiments

1. **The Milgram Experiment:** A study on obedience to authority figures, conducted by Stanley Milgram in the 1960s.
2. **The Stanford Prison Experiment:** A study of the psychological effects of becoming a prisoner or prison guard, conducted by Philip Zimbardo in 1971.
3. **The Asch Conformity Experiment:** A study on the effects of group pressure on individual behavior, conducted by Solomon Asch in the 1950s.
4. **The Pavlov's Dog Experiment:** A study on classical conditioning, conducted by Ivan Pavlov in the 1890s.
5. **The Marshmallow Test:** A study on delayed gratification, conducted by Walter Mischel in the 1960s.
6. **The Hawthorne Effect:** A study on the effects of changes in working conditions on productivity, conducted by Elton Mayo in the 1920s.
7. **The Bobo Doll Experiment:** A study on the effects of observing aggressive behavior, conducted by Albert Bandura in the 1960s.
8. **The Bystander Effect:** A study on the diffusion of responsibility, conducted by John Darley and Bibb Latané in the 1960s.
9. **The Heinz Dilemma:** A study on moral development, conducted by Lawrence Kohlberg in the 1950s.
10. **The Trolley Problem:** A thought experiment in ethics, first introduced by Philippa Foot in 1967.

Calibration of Analytical Equipment and Apparatus

Calibration is the process of verifying the accuracy of a measuring instrument by comparing its readings with those of a known standard. This is important for ensuring the accuracy and reliability of analytical results.

1. **Purpose of Calibration:** The purpose of calibration is to ensure that the readings obtained from analytical equipment and apparatus are accurate and reliable. This is important for achieving consistent and reproducible results in analytical measurements.
2. **Types of Calibration:** There are two main types of calibration: external and internal. External calibration involves comparing the readings of the instrument with those of a known standard, while internal calibration involves adjusting the instrument's internal settings to ensure accuracy.
3. **Calibration Standards:** Calibration standards are materials or devices with known properties that are used to verify the accuracy of analytical equipment and apparatus. These standards can be physical objects, such as weights or rulers, or chemical substances with known concentrations.
4. **Calibration Procedures:** Calibration procedures vary depending on the type of equipment or apparatus being calibrated. Common procedures include zeroing, span calibration, and linearity checks.
5. **Calibration Frequency:** The frequency of calibration depends on several factors, including the type of equipment or apparatus, its usage, and the requirements of the analytical method. Some equipment may require daily calibration, while others may only need to be calibrated once a year.
6. **Documentation:** Proper documentation of calibration procedures and results is important for ensuring the accuracy and reliability of analytical results. This includes recording the date and time of calibration, the standards used, and any adjustments made to the equipment or apparatus.
7. **Quality Control:** Calibration is an important part of quality control in analytical measurements. Regular calibration helps to identify and correct any issues with the equipment or apparatus, ensuring that analytical results are accurate and reliable.

In summary, calibration is an essential process for ensuring the accuracy and reliability of analytical equipment and apparatus. It involves comparing the readings of the instrument with those of a known standard, and making any necessary adjustments to ensure accuracy. Proper documentation and regular calibration are important for maintaining quality control in analytical measurements.

Determination of Hardness of Water Sample by EDTA Method

The hardness of water is due to the presence of calcium and magnesium salts in water. The EDTA method is one of the three main methods for determining the hardness of water. EDTA stands for Ethylenediaminetetraacetic acid. This reagent can form EDTA-metal complexes by reacting with metal ions, except for alkali metal ions.

Procedure for Calculation of Hardness of Water by EDTA Titration

1. Take a sample volume of 20ml (V ml).
2. Dilute 20ml of the sample in an Erlenmeyer flask to 40ml by adding 20ml of distilled water.
3. Add 1 mL of ammonia buffer to bring the pH to 10 ± 0.1 .
4. Add 1 or 2 drops of the indicator solution.
5. Add EDTA titrant to the sample with vigorous shaking until the wine red color just turns blue.

Calculation of Hardness

Hardness (EDTA), as $\text{mg/L} = \frac{A}{S} \times T \times 1,000$ where A = mL of EDTA titrant used, T = Titer of EDTA titrant, mg CaCO_3 per mL of EDTA titrant, and S = mL of sample volume. Report as “Total Hardness (EDTA) = _____ mg/L as CaCO_3 ” or as “Calcium Hardness (EDTA) = _____ mg/L as CaCO_3 ”.

Determination of Alkalinity of Water Sample

Alkalinity is a measure of the ability of water to neutralize acids. It is an important parameter in determining the quality of water for various uses. Here are the steps to determine the alkalinity of a water sample:

1. Collect a water sample in a clean container.
2. Measure the pH of the water sample using a pH meter or pH test strips.
3. If the pH is above 7, the water is alkaline.
4. To determine the exact alkalinity, titrate the water sample with a standard acid solution, such as sulfuric acid or hydrochloric acid, using a burette.
5. The endpoint of the titration is reached when the pH of the water sample drops to a value between 4.3 and 4.5.
6. Calculate the alkalinity of the water sample using the volume of the acid solution used and the concentration of the acid solution.

These are the basic steps to determine the alkalinity of a water sample. It is important to follow proper laboratory procedures and safety guidelines when performing this test.

4. Determination of pH by titrimetric method

Titrimetric method is a common laboratory technique used to determine the pH of a solution. The process involves the following steps:

1. A solution of known concentration, called the titrant, is prepared. This is usually a strong acid or base.
2. The solution to be tested, called the analyte, is measured and placed in a container.
3. A few drops of an indicator are added to the analyte. The indicator is a substance that changes color depending on the pH of the solution.
4. The titrant is slowly added to the analyte while stirring. The indicator changes color when the pH of the solution reaches a certain value.
5. The volume of titrant added is recorded when the color change occurs. This is called the endpoint.
6. The pH of the solution can be calculated using the volume of titrant added and the concentration of the titrant.

This method is accurate and reliable, but it requires careful preparation and execution. It is commonly used in laboratories and in industrial processes where precise pH measurements are required.

Determination of Surface Tension of Given Liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the liquid molecules. It is defined as the force acting per unit length on the surface of the liquid. There are several methods to determine the surface tension of a given liquid, including:

1. **Capillary rise method:** This method involves measuring the height to which a liquid rises in a capillary tube due to the surface tension of the liquid. The surface tension is then calculated using the formula: $T = \frac{h \rho g r}{2 \cos \theta}$, where T is the surface tension, h is the height of the liquid rise, ρ is the density of the liquid, g is the acceleration due to gravity, r is the radius of the capillary tube, and θ is the contact angle between the liquid and the tube.
2. **Maximum bubble pressure method:** This method involves measuring the maximum pressure required to blow a bubble from the end of a capillary tube immersed in the liquid. The surface tension is then calculated using the formula: $T = \frac{P_m R}{3}$, where T is the surface tension, P_m is the maximum pressure, and R is the radius of the capillary tube.
3. **Drop weight method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a capillary tube. The surface tension is then calculated using the formula: $T = \frac{mg}{d}$, where T is the surface tension, m is the mass of the droplet, g is the acceleration due to gravity, and d is the diameter of the capillary tube.

4. **Wilhelmy plate method:** This method involves immersing a thin, flat plate into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: $T = F/L$, where T is the surface tension, F is the force required to detach the plate, and L is the perimeter of the plate.
5. **Du Nouy ring method:** This method involves immersing a ring into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: $T = (F - F_c)/2R$, where T is the surface tension, F is the force required to detach the ring, F_c is the correction factor, and R is the radius of the ring.

These are some of the common methods used to determine the surface tension of a given liquid. The choice of method depends on the specific requirements and conditions of the experiment.

Determination of Viscosity of a given liquid by viscometer

1. Viscosity is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate in a fluid.
2. A viscometer is an instrument used to measure the viscosity of a fluid.
3. There are several types of viscometers, including capillary, rotational, and falling ball viscometers.
4. To determine the viscosity of a given liquid using a viscometer, the liquid is placed in the instrument and its flow rate is measured.
5. The flow rate is then used to calculate the viscosity of the liquid using the appropriate formula for the type of viscometer being used.
6. The viscosity of a liquid can be affected by factors such as temperature and pressure, so it is important to control these variables during the measurement.
7. The viscosity of a liquid can also be affected by the presence of impurities or dissolved substances, so it is important to use a pure sample of the liquid for the measurement.
8. The viscosity of a liquid is an important property that can affect its behavior in various applications, such as lubrication, flow through pipes, and mixing.

Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. Its chemical formula is $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$.

3. It is commonly used as a standard solution in redox titrations due to its stability and the ease with which it can be prepared.
4. To determine the strength of a ferrous ammonium sulfate solution, a redox titration can be performed using an external indicator.
5. An external indicator is a substance that changes color at the endpoint of the titration, indicating that the reaction is complete.
6. Common external indicators used in redox titrations include diphenylamine, starch, and potassium ferricyanide.
7. The choice of external indicator depends on the specific redox reaction being studied.
8. The procedure for determining the strength of a ferrous ammonium sulfate solution using an external indicator involves adding the indicator to the solution and titrating it against a standard solution of a known oxidizing agent.
9. The volume of the oxidizing agent required to reach the endpoint, as indicated by the color change of the external indicator, can be used to calculate the strength of the ferrous ammonium sulfate solution.
10. It is important to carefully control the experimental conditions, such as the temperature and pH, to ensure accurate results.

Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is a primary standard substance that can be used to determine the strength of a solution.
2. An internal indicator is used to determine the end point of the titration.
3. The internal indicator used in this experiment is potassium iodide.
4. The reaction between potassium dichromate and potassium iodide produces iodine, which can be observed as a color change.
5. The iodine produced is then titrated with sodium thiosulfate until the solution becomes colorless, indicating the end point of the titration.
6. From the volume of sodium thiosulfate used, the strength of the potassium dichromate solution can be calculated.

Determination of Available Chlorine in Bleaching Powder

1. Bleaching powder is a white powder that is mainly used as a disinfectant and as a bleaching agent.
2. It is also known as calcium hypochlorite and its chemical formula is $\text{Ca}(\text{OCl})_2$.
3. The effectiveness of bleaching powder depends on the amount of available chlorine it contains.

4. Available chlorine is the amount of chlorine that is available to react with other substances.
5. The determination of available chlorine in bleaching powder is important to ensure its effectiveness as a disinfectant and as a bleaching agent.
6. The available chlorine in bleaching powder can be determined by titrating a known amount of bleaching powder with a standard solution of sodium thiosulfate.
7. The reaction between sodium thiosulfate and available chlorine produces iodine, which can be titrated with a standard solution of sodium thiosulfate.
8. The amount of sodium thiosulfate used in the titration is used to calculate the amount of available chlorine in the bleaching powder.
9. The result is expressed as a percentage of available chlorine in the bleaching powder.

Determination of Chloride Content in Water Sample

1. Chloride ions are one of the major inorganic anions in water and wastewater.
2. The chloride test measures the level of chloride ions in water.
3. The presence of chloride ions can be an indicator of water pollution from sources such as industrial waste, sewage, and road salt.
4. The most common method for determining the chloride content in water is the argentometric method, also known as the Mohr method.
5. In this method, a measured volume of water sample is titrated with a standard solution of silver nitrate using potassium chromate as an indicator.
6. The end point of the titration is indicated by the formation of a red-brown precipitate of silver chromate.
7. The amount of silver nitrate used in the titration is used to calculate the concentration of chloride ions in the water sample.
8. Other methods for determining the chloride content in water include ion chromatography, ion-selective electrodes, and spectrophotometry.
9. The acceptable level of chloride ions in drinking water varies by location and is regulated by local authorities.
10. High levels of chloride ions in water can have negative effects on the taste and odor of the water, as well as on the environment and aquatic life.

Preparation of Phenol Formaldehyde (PF) Resin

Phenol Formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol with formaldehyde. Here are the steps involved in the preparation of PF resin:

1. **Phenol and Formaldehyde Reaction:** The first step in the preparation of PF resin is the reaction of phenol with formaldehyde in the presence of an acidic or basic catalyst. The ratio of formaldehyde to phenol is typically between 1.5:1 and 2:1. The reaction is carried out at a temperature of around 90°C.
2. **Resin Formation:** The reaction between phenol and formaldehyde produces a mixture of low molecular weight pre-polymers, which then undergo further condensation reactions to form the resin. The resin formation process can be controlled by adjusting the reaction conditions, such as temperature, catalyst concentration, and reaction time.
3. **Resin Curing:** The final step in the preparation of PF resin is the curing process, where the resin is heated to a high temperature to complete the polymerization process. The curing temperature and time depend on the specific application of the resin.

PF resin is widely used in the production of various products, including plywood, laminates, and molding compounds, due to its high strength, heat resistance, and chemical resistance. It is also used as an adhesive and a binding agent in various industrial applications.

12. Preparation of Urea Formaldehyde (UF) Resin

Urea Formaldehyde (UF) resin is a type of thermosetting polymer that is commonly used in the production of adhesives, finishes, and molded objects. It is made by reacting urea and formaldehyde in the presence of a catalyst.

The steps involved in the preparation of UF resin are as follows:

1. **Urea and formaldehyde are mixed:** The first step in the preparation of UF resin is to mix urea and formaldehyde in the desired ratio. The ratio of urea to formaldehyde is typically between 1:1.5 and 1:2.
2. **A catalyst is added:** A catalyst is added to the mixture to speed up the reaction between urea and formaldehyde. Common catalysts used in the preparation of UF resin include ammonia, sulfuric acid, and hydrochloric acid.
3. **The mixture is heated:** The mixture is heated to a temperature of around 90-100°C to facilitate the reaction between urea and formaldehyde.
4. **The reaction is allowed to proceed:** The reaction between urea and formaldehyde is allowed to proceed for a period of time, typically between 30 minutes and 2 hours.
5. **The resin is cooled and neutralized:** After the reaction is complete, the resin is cooled and neutralized to remove any excess acid or base.

6. **The resin is concentrated:** The resin is then concentrated by removing excess water to increase its solids content.
7. **The resin is ready for use:** The resulting UF resin is ready for use in the production of adhesives, finishes, and molded objects.

Preparation of Adipic Acid and Paracetamol

Adipic acid and paracetamol are two important chemicals that have a wide range of applications. Here is an overview of their preparation methods:

Adipic Acid

Adipic acid is an organic compound with the formula $(\text{CH}_2)_4(\text{COOH})_2$. It is a white crystalline powder that is mainly used as a precursor to the production of nylon. Here are the steps involved in its preparation:

1. The most common method for the production of adipic acid is the oxidation of cyclohexane. This process involves the use of a catalyst, such as cobalt or manganese, to facilitate the reaction.
2. Another method for the production of adipic acid is the oxidation of cyclohexanol or cyclohexanone. This process is similar to the oxidation of cyclohexane, but it involves the use of different starting materials.
3. Adipic acid can also be produced from renewable resources, such as sugar or corn starch, through a fermentation process. This method is still in the development stage and is not yet widely used.

Paracetamol

Paracetamol, also known as acetaminophen, is a widely used pain reliever and fever reducer. It is commonly found in over-the-counter medications, such as Tylenol. Here are the steps involved in its preparation:

1. The most common method for the production of paracetamol is the nitration of phenol. This process involves the reaction of phenol with nitric acid to produce a mixture of nitrophenols.
2. The mixture of nitrophenols is then separated and the desired isomer, para-nitrophenol, is isolated.
3. The para-nitrophenol is then reduced to para-aminophenol using a reducing agent, such as tin and hydrochloric acid.
4. Finally, the para-aminophenol is acetylated using acetic anhydride to produce paracetamol.

These are the common methods for the preparation of adipic acid and paracetamol. Both chemicals have important applications in various industries and are widely used in the production of various products.

14. Determination of Cell Conductance of a solution

Cell conductance is a measure of a solution's ability to conduct an electric current. It is determined by the concentration and mobility of ions in the solution. Here are the steps to determine the cell conductance of a solution:

1. Prepare the solution to be tested by dissolving a known amount of solute in a known volume of solvent.
2. Set up the conductivity cell by connecting it to a conductivity meter.
3. Rinse the conductivity cell with distilled water and dry it.
4. Fill the conductivity cell with the solution to be tested.
5. Turn on the conductivity meter and wait for the reading to stabilize.
6. Record the conductivity reading displayed on the meter.
7. Repeat the measurement with different concentrations of the solution to obtain a calibration curve.
8. Use the calibration curve to determine the cell conductance of the solution.

It is important to note that the temperature of the solution can affect its conductivity, so it is recommended to perform the measurement at a constant temperature. Additionally, the conductivity cell should be calibrated regularly to ensure accurate measurements.

Determination of Rate constant of hydrolysis of esters

1. The rate constant of hydrolysis of esters can be determined by measuring the rate of reaction at different concentrations of the reactants.
2. The reaction between an ester and water to produce a carboxylic acid and an alcohol is known as hydrolysis.
3. The rate of hydrolysis of an ester can be determined by measuring the change in concentration of one of the reactants or products over time.
4. The rate law for the hydrolysis of an ester can be written as: $\text{rate} = k[\text{ester}][\text{water}]$, where k is the rate constant and $[\text{ester}]$ and $[\text{water}]$ are the concentrations of the ester and water, respectively.
5. The rate constant, k , can be determined by measuring the initial rate of the reaction at different initial concentrations of the ester and water and plotting the data on a graph.
6. The slope of the line obtained from the graph will give the value of the rate constant, k .
7. The units of the rate constant will depend on the order of the reaction with respect to the ester and water.
8. The order of the reaction can be determined by varying the concentration of one of the reactants while keeping the concentration of the other reactant constant and observing the effect on the rate of the reaction.

9. The hydrolysis of esters is typically a first-order reaction with respect to the ester and zero-order with respect to water.
10. The rate constant of the hydrolysis of an ester can also be determined by measuring the change in pH of the reaction mixture over time, as the hydrolysis of an ester produces a carboxylic acid, which will affect the pH of the solution.
11. The rate constant can also be determined by measuring the change in conductivity of the reaction mixture over time, as the hydrolysis of an ester produces ions, which will affect the conductivity of the solution.
12. The rate constant of the hydrolysis of an ester can also be determined by measuring the change in absorbance of the reaction mixture over time, if one of the reactants or products has a characteristic absorbance in the UV-visible region.
13. The rate constant can also be determined by measuring the change in volume of the reaction mixture over time, as the hydrolysis of an ester produces a carboxylic acid and an alcohol, which may have different densities than the ester and water.
14. The rate constant of the hydrolysis of an ester can also be determined by measuring the change in mass of the reaction mixture over time, as the hydrolysis of an ester produces a carboxylic acid and an alcohol, which may have different molecular weights than the ester and water.
15. The rate constant of the hydrolysis of an ester can also be determined by measuring the change in temperature of the reaction mixture over time, as the hydrolysis of an ester is an exothermic reaction and will release heat.

Element detection and identification of functional groups in organic compounds

1. **Element detection** refers to the process of identifying the presence of specific elements in a given compound.
2. **Functional groups** are specific groups of atoms within a molecule that are responsible for the characteristic chemical reactions of that molecule.
3. There are several methods for detecting elements and identifying functional groups in organic compounds, including:
 - **Combustion analysis:** This method involves burning a sample of the compound and analyzing the resulting products to determine the presence of elements such as carbon, hydrogen, and nitrogen.
 - **Infrared spectroscopy:** This technique uses infrared radiation to identify the presence of specific functional groups based on their characteristic absorption patterns.
 - **Mass spectrometry:** This method involves ionizing a sample of the compound and analyzing the resulting fragments to determine the presence of specific elements and functional groups.
 - **Nuclear magnetic resonance spectroscopy:** This technique uses

the interaction of nuclear spins with a magnetic field to identify the presence of specific functional groups based on their characteristic chemical shifts.

4. These methods can be used in combination to provide a comprehensive analysis of the elements and functional groups present in a given organic compound.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose any 10 experiments from the list provided.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read the list of experiments and be prepared for any changes made by the instructor.
- Students should also communicate with the instructor if they have any questions or concerns about the experiments chosen.

Course Outcomes

Course outcomes are statements that describe the knowledge, skills, and abilities that students are expected to demonstrate upon completion of a course. These outcomes serve as a guide for instructors in designing course content and assessments, and for students in understanding the goals of the course.

Some key points to consider when developing course outcomes include:

1. Outcomes should be specific, measurable, and achievable within the scope of the course.
2. Outcomes should be aligned with the overall goals of the program or curriculum.
3. Outcomes should be written in clear, concise language that is easily understood by students.
4. Outcomes should focus on what students will be able to do, rather than what they will know.
5. Outcomes should be reviewed and revised regularly to ensure they remain relevant and up-to-date.

In summary, course outcomes provide a clear and concise roadmap for both instructors and students, helping to ensure that the course is effective in achieving

its intended goals. By carefully developing and regularly reviewing course outcomes, instructors can create a more engaging and effective learning experience for their students.

Upon completion of the course the student should be able to:

- Demonstrate an understanding of the fundamental concepts and principles of the subject matter.
- Apply the knowledge and skills acquired in the course to solve problems and make informed decisions.
- Communicate effectively using the language and terminology of the subject matter.
- Work effectively both independently and as part of a team.
- Demonstrate critical thinking and analytical skills.
- Demonstrate an awareness of ethical and social issues related to the subject matter.
- Demonstrate an ability to engage in lifelong learning and professional development.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should possess upon completion of a course. These outcomes are typically aligned with the course objectives and are used to assess student learning.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. The taxonomy consists of six levels, with each level building upon the previous one. These levels are:

1. **Remembering:** The ability to recall or retrieve previously learned information.
2. **Understanding:** The ability to comprehend the meaning of the material.
3. **Applying:** The ability to use learned material in new and concrete situations.
4. **Analyzing:** The ability to break down material into its component parts so that its organizational structure may be understood.
5. **Evaluating:** The ability to make judgments about the value of ideas or materials.
6. **Creating:** The ability to put parts together to form a new whole.

Course outcomes can be written using Bloom's Taxonomy to ensure that they are specific, measurable, and achievable. For example, a course outcome for a biology course might be: "Students will be able to analyze the structure and

function of the major organelles in a eukaryotic cell.” This outcome aligns with the fourth level of Bloom’s Taxonomy, Analyzing.

In summary, course outcomes are statements that describe what students should be able to do upon completion of a course, and Bloom’s Taxonomy is a framework that can be used to write specific, measurable, and achievable course outcomes.

Level

1. A level is a tool used to determine if a surface is horizontal or vertical.
2. It consists of a small glass tube filled with liquid, usually alcohol or a colored spirit, with an air bubble inside.
3. When the level is placed on a surface, the bubble will move to the highest point of the tube, indicating if the surface is level or not.
4. There are different types of levels, including spirit levels, laser levels, and water levels.
5. Levels are commonly used in construction, carpentry, and surveying, among other fields.
6. It is important to use a level when installing objects such as shelves or cabinets to ensure they are straight and level.
7. Levels can also be used to determine the slope or grade of a surface, which is useful in landscaping and road construction.
8. Some levels come with additional features such as rulers or digital displays for increased accuracy and ease of use.

CO-1 Get an understanding of the use of different analytical instruments. K3

Analytical instruments are tools used to measure, analyze, and quantify the physical and chemical properties of substances. These instruments are used in a variety of fields, including chemistry, biology, physics, and engineering. Some common analytical instruments include:

1. **Spectrophotometers:** These instruments measure the absorption or transmission of light by a sample. They are commonly used to determine the concentration of a substance in a solution.
2. **Chromatographs:** These instruments separate mixtures into their individual components. They are commonly used to identify and quantify the components of a mixture.
3. **Mass spectrometers:** These instruments measure the mass-to-charge ratio of ions. They are commonly used to identify the chemical composition of a sample.

4. **Electrochemical instruments:** These instruments measure the electrical properties of a sample. They are commonly used to determine the concentration of ions in a solution.
5. **Thermal analyzers:** These instruments measure the thermal properties of a sample. They are commonly used to determine the melting point, boiling point, and heat capacity of a substance.

Understanding the use of these different analytical instruments is important for conducting accurate and precise measurements in various fields of study. By selecting the appropriate instrument for a given task, researchers can obtain reliable data to support their conclusions.

CO₂: Measuring Molecular/System Properties such as Surface Tension

Carbon dioxide (CO₂) is a colorless, odorless gas that is naturally present in the Earth's atmosphere. It is an important greenhouse gas that helps regulate the temperature of the planet. However, human activities such as burning fossil fuels and deforestation have increased the concentration of CO₂ in the atmosphere, contributing to global warming.

Surface tension is a property of liquids that arises from the cohesive forces between molecules at the surface of a liquid. It is the force that allows insects to walk on water and causes water droplets to form beads on a surface.

To measure the surface tension of CO₂, several methods can be used, including:

1. **Maximum Bubble Pressure Method:** This method involves measuring the maximum pressure required to force a gas bubble through a liquid.
2. **Drop Weight Method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube.
3. **Wilhelmy Plate Method:** This method involves measuring the force required to detach a thin plate from the surface of a liquid.

These methods can be used to measure the surface tension of CO₂ and other gases, providing valuable information about their molecular and system properties.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- The viscosity of formation water is a function of pressure, temperature, and dissolved solids.

- In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature.
- Dissolved gas in the formation water at reservoir conditions generally results in a negligible effect on water viscosity.

Conductance of Solution

- Conductivity is a measure of a solution's ability to conduct an electric current.
- High quality deionized water has a conductivity of about 0.05 S/cm at 25 °C, typical drinking water is in the range of 200–800 S/cm, while sea water is about 50 mS/cm (or 0.05 S/cm).
- The more chloride ions present in the water, the higher the conductivity.
- In general, the conductivity of waterways in the United States varies from 50 to 1500 μ mhos/cm, and inland freshwater lake studies reveal a conductivity of 150 to 500 μ mhos/cm.

Chloride and Iron Content in the Water

- Chloride ions are approximately 1.9 percent of the mass of seawater.
- The presence of chloride ions in water can affect its conductivity, as mentioned above.

K3

K3 is a term that can refer to several different things. Without more context, it is difficult to determine which specific topic you are referring to. Some possible interpretations of K3 include:

1. K3, a Belgian-Dutch girl group formed in 1998.
2. K3, a mountain in the Karakoram range, also known as Broad Peak.
3. K3, a type of visa for spouses of U.S. citizens to enter the United States while waiting for their immigrant visa to be processed.
4. K3, a type of South Korean high-speed train.
5. K3, a type of tax form used in the United States for reporting income from partnerships, S corporations, and trusts.

CO-3 Measure the hardness and alkalinity of the water. K3

Hardness and alkalinity are two important parameters of water quality. Hardness refers to the concentration of calcium and magnesium ions in water, while alkalinity refers to the ability of water to neutralize acids.

Measuring Hardness

1. One way to measure the hardness of water is by using a titration method. This involves adding a chemical reagent to a water sample until a color change occurs, indicating that all the calcium and magnesium ions have reacted with the reagent.
2. Another way to measure hardness is by using a test strip or a digital meter. These methods are quicker and easier to use than titration, but may not be as accurate.

Measuring Alkalinity

1. Alkalinity can be measured using a titration method similar to the one used for measuring hardness. A chemical reagent is added to the water sample until a color change occurs, indicating that all the acid-neutralizing compounds in the water have reacted with the reagent.
2. Another way to measure alkalinity is by using a pH meter. The pH of the water sample is measured before and after the addition of a known amount of acid. The difference in pH can be used to calculate the alkalinity of the water.

It is important to measure the hardness and alkalinity of water to ensure that it is suitable for its intended use. For example, water with high hardness can cause scaling in pipes and appliances, while water with low alkalinity can be corrosive.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound that can be prepared through several methods. Some of the most common methods of preparation of phenol are:

1. **From Haloarenes:** Chlorobenzene is an example of a haloarene which is formed by the monosubstitution of the benzene ring. When chlorobenzene is fused with sodium hydroxide at 623K and 320 atm, sodium phenoxide is produced .
2. **From Diazonium Salts:** Phenols can also be prepared from diazonium salts.
3. **From Benzene Sulphonic Acid:** Sodium phenoxide is produced by fusing the sodium salt of benzene sulphonic acid with sodium hydroxide at 573 – 623K. Sodium phenoxide is converted to phenol when it is treated with dilute acid or carbon dioxide.
4. **From Cumene:** Phenol can also be prepared by the air oxidation of cumene.

5. **Industrial Preparation:** Phenol is extracted from the middle oil fraction for commercial purposes at 443 – 503K. Another industrial method is the Raschig’s Process, which involves the reaction of benzene, hydrogen chloride, and air at 523K.

Formaldehyde & Urea Formaldehyde Resin, Adipic Acid and Paracetamol

Formaldehyde

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- It is used in pressed-wood products, such as particleboard, plywood, and fiberboard; glues and adhesives; permanent-press fabrics; paper product coatings; and certain insulation materials.

Urea Formaldehyde Resin

- Urea-formaldehyde (UF), also known as urea-methanal, is a nontransparent thermosetting resin or polymer.
- It is produced from urea and formaldehyde.
- These resins are used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- Using formaldehyde and urea as raw materials, a stable urea-formaldehyde resin (UF) is synthesized by the “alkali-acid-alkali” method.
- Unlike most thermosetting resins, UF often shows the appearance of crystal domains.

Adipic Acid

- Melamine, polyvinyl alcohol, and adipic acid dihydrazide as modifiers were added to urea-formaldehyde (UF) resin for reducing the free formaldehyde.
- The influence of addition amount and addition stage of modifiers on the physicochemical property and free formaldehyde content of UF resin was tested.

Paracetamol

K3

- K3 is a synthetic, artificially produced form of vitamin K that doesn’t occur naturally .
- This is unlike the other two forms of vitamin K — vitamin K1, known as phylloquinone, and vitamin K2, called menaquinone .
- Vitamin K3 may be converted into K2 in your liver .

- Vitamin K3 is also known as menadione .
- Vitamin K is important for blood clotting and bone health .
- Although vitamin K3 has demonstrated anticancer and antibacterial properties in test-tube studies, it has potential side effects .
- K3 is also the name of a form used by the IRS, called Schedule K-3 (Form 1065) .

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by the symbol k , is a measure of the speed of a chemical reaction. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers, as given by the rate law of the reaction.

To estimate the rate constant of a reaction, the following steps can be followed:

1. Determine the order of the reaction with respect to each reactant by conducting experiments and analyzing the data.
2. Write the rate law for the reaction using the determined orders of the reactants.
3. Conduct experiments to measure the rate of the reaction at different concentrations of the reactants.
4. Use the data obtained from the experiments to calculate the value of the rate constant k by substituting the values of the rate and the concentrations of the reactants into the rate law.

It is important to note that the value of the rate constant k is dependent on the temperature at which the reaction is conducted. Therefore, the temperature must be kept constant during the experiments to obtain accurate results. Additionally, the units of the rate constant k will depend on the overall order of the reaction. For example, for a first-order reaction, the units of k will be s^{-1} , while for a second-order reaction, the units will be $M^{-1} s^{-1}$.

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1. Page 32 is the thirty-second page of a forty-page document.
2. It is located after page 31 and before page 33.
3. The content of page 32 will vary depending on the document it is a part of.
4. It is important to read and understand the content of page 32 in the context of the entire document.
5. Page 32 may contain text, images, tables, or other types of information.
6. It is important to review page 32 carefully to ensure that the information is accurate and relevant.
7. If studying for an exam, it may be helpful to take notes on the content of page 32 to aid in retention and understanding.

ENGINEERING MATHEMATICS LAB

Engineering Mathematics Lab is a course designed to provide students with hands-on experience in applying mathematical concepts and techniques to solve engineering problems. The course typically includes the following topics:

1. **Numerical Methods:** Students learn various numerical methods for solving mathematical problems, including root-finding, interpolation, numerical integration, and numerical differentiation.
2. **Linear Algebra:** Students learn the basics of linear algebra, including matrix operations, determinants, and eigenvalues and eigenvectors. They also learn how to use linear algebra to solve systems of linear equations.
3. **Differential Equations:** Students learn how to solve ordinary differential equations (ODEs) and partial differential equations (PDEs) using analytical and numerical methods.
4. **Probability and Statistics:** Students learn the basics of probability and statistics, including probability distributions, descriptive statistics, and hypothesis testing. They also learn how to use statistical methods to analyze data.
5. **Optimization:** Students learn various optimization techniques, including linear programming, nonlinear programming, and dynamic programming. They also learn how to use optimization methods to solve engineering problems.

In the Engineering Mathematics Lab, students work on projects and assignments that require them to apply the mathematical concepts and techniques they have learned to solve real-world engineering problems. The course is typically taken by engineering students in their second or third year of study.

Course Objectives:

1. To provide students with a comprehensive understanding of the subject matter.
2. To equip students with the necessary skills and knowledge to apply the concepts learned in real-life situations.
3. To encourage critical thinking and problem-solving skills.
4. To foster a deep appreciation for the subject and its relevance in the world.
5. To prepare students for further study or career opportunities in the field.
6. To promote active engagement and participation in the learning process.
7. To facilitate the development of effective communication and collaboration skills.
8. To provide a supportive and inclusive learning environment that caters to diverse learning needs.
9. To assess student progress and provide timely feedback to support their learning journey.

10. To encourage lifelong learning and continuous professional development.

1. To enable the students to understand about the fundamental concepts of analytical instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are commonly used in scientific research, medical diagnosis, and industrial processes.

Some fundamental concepts of analytical instruments include:

- **Accuracy:** The degree to which the measurement result agrees with the true value of the quantity being measured.
- **Precision:** The degree to which repeated measurements of the same quantity agree with each other.
- **Sensitivity:** The smallest change in the quantity being measured that can be detected by the instrument.
- **Resolution:** The smallest difference between two measurements that can be distinguished by the instrument.
- **Calibration:** The process of adjusting the instrument to ensure that its measurements are accurate.
- **Signal-to-noise ratio:** The ratio of the strength of the signal being measured to the background noise.

Understanding these concepts is essential for the proper use and interpretation of analytical instruments. It is important for students to learn about these concepts in order to effectively use analytical instruments in their future studies and careers.

Analysis of Chloride Content, Hardness, and Alkalinity of Water

1. **Chloride Content:** Chloride content in water is an important parameter to measure as it can indicate the presence of pollution from human or animal waste, industrial effluents, or saltwater intrusion. The chloride content of water can be measured using various methods, including titration with silver nitrate, ion chromatography, and ion-selective electrodes.
2. **Hardness:** Hardness in water is caused by the presence of dissolved calcium and magnesium ions. It can cause scaling in pipes and appliances, and can affect the taste and appearance of water. Hardness can be measured using titration with ethylenediaminetetraacetic acid (EDTA) or by using a colorimetric test kit.

3. **Alkalinity:** Alkalinity is a measure of the water's ability to neutralize acids. It is important to measure alkalinity as it can affect the pH of the water and its ability to support aquatic life. Alkalinity can be measured using titration with a strong acid, such as sulfuric acid, or by using a colorimetric test kit.

These are some of the key parameters that are important to understand when analyzing the quality of water. By measuring the chloride content, hardness, and alkalinity of water, we can gain a better understanding of its overall quality and suitability for various uses.

3. To enable the students to understand about the measure of pH, surface tension and viscosity of a liquid.

- **pH** is a measure of the acidity or basicity of a solution. It is defined as the negative logarithm of the hydrogen ion concentration in a solution. The pH scale ranges from 0 to 14, with 7 being neutral. A pH less than 7 indicates an acidic solution, while a pH greater than 7 indicates a basic solution. The pH can be measured using a pH meter or by using pH indicator paper.
- **Surface tension** is the property of a liquid that allows it to resist an external force. It is caused by the cohesive forces between the molecules of the liquid. Surface tension can be measured using a number of methods, including the drop weight method, the maximum bubble pressure method, and the Wilhelmy plate method.
- **Viscosity** is a measure of a fluid's resistance to flow. It is defined as the ratio of the shear stress to the shear rate in a fluid. The viscosity of a fluid can be measured using a number of methods, including the capillary tube method, the falling ball method, and the rotational viscometer method.

These properties of liquids are important in many fields, including chemistry, physics, and engineering. Understanding how to measure and interpret these properties can help students better understand the behavior of liquids in various situations.

4. To enable the students to understand about the preparation of different resins

Resins are solid or highly viscous substances that are typically convertible into polymers. They are usually mixtures of organic compounds, mainly terpenes and their derivatives. Here are some points to consider when preparing different resins:

1. **Types of resins:** There are several types of resins, including natural resins, synthetic resins, and modified natural resins. Each type of resin has its own unique properties and preparation methods.
2. **Raw materials:** The raw materials used in the preparation of resins vary depending on the type of resin being prepared. For example, natural resins are obtained from plants, while synthetic resins are prepared from chemicals.
3. **Preparation methods:** The methods used to prepare resins vary depending on the type of resin being prepared. Some common methods include polymerization, condensation, and addition reactions.
4. **Safety precautions:** The preparation of resins often involves the use of chemicals and high temperatures. It is important to follow safety precautions, such as wearing protective clothing and working in a well-ventilated area.
5. **Quality control:** The quality of the resins prepared should be monitored to ensure that they meet the desired specifications. This can be done through various tests, such as viscosity and hardness tests.

In conclusion, the preparation of different resins involves several steps, including selecting the appropriate type of resin, obtaining the raw materials, using the appropriate preparation method, following safety precautions, and monitoring the quality of the final product. It is important to have a good understanding of these steps to successfully prepare different resins.

5. Synthesis of Organic Compounds: Adipic Acid and Paracetamol by Conventional and Green Route

Adipic acid and paracetamol are two organic compounds that can be synthesized through both conventional and green routes. Here, we will discuss the differences between these two methods of synthesis.

Adipic Acid

Adipic acid is an important dicarboxylic acid that is used in the production of nylon. It can be synthesized through both conventional and green routes.

Conventional Route

The conventional route for the synthesis of adipic acid involves the oxidation of cyclohexane with nitric acid. This process produces nitrous oxide, a greenhouse gas, as a byproduct.

Green Route

The green route for the synthesis of adipic acid involves the oxidation of cyclohexanol or cyclohexanone with hydrogen peroxide. This process produces water as the only byproduct, making it a more environmentally friendly option.

Paracetamol

Paracetamol is a widely used analgesic and antipyretic drug. It can also be synthesized through both conventional and green routes.

Conventional Route

The conventional route for the synthesis of paracetamol involves the nitration of phenol, followed by the reduction of the nitro group to an amino group, and then acetylation.

Green Route

The green route for the synthesis of paracetamol involves the direct acetylation of para-aminophenol with acetic anhydride. This process produces fewer byproducts and is considered to be more environmentally friendly.

In conclusion, the synthesis of organic compounds such as adipic acid and paracetamol can be carried out through both conventional and green routes. The green routes are generally considered to be more environmentally friendly due to the production of fewer byproducts. It is important for students to understand the differences between these two methods of synthesis and the impact they have on the environment.

LIST OF EXPERIMENTS

1. **Milgram Experiment:** A study on obedience to authority figures, conducted by Stanley Milgram in the 1960s.
2. **Stanford Prison Experiment:** A study of the psychological effects of becoming a prisoner or prison guard, conducted by Philip Zimbardo in 1971.
3. **Pavlov's Dogs:** A study on classical conditioning, conducted by Ivan Pavlov in the 1890s.
4. **Asch Conformity Experiment:** A study on conformity, conducted by Solomon Asch in the 1950s.
5. **Hawthorne Effect:** A study on the effect of changes in working conditions on productivity, conducted by Elton Mayo in the 1920s.
6. **Bobo Doll Experiment:** A study on the effects of observing aggressive behavior, conducted by Albert Bandura in the 1960s.
7. **Little Albert Experiment:** A study on classical conditioning of fear, conducted by John B. Watson and Rosalie Rayner in 1920.
8. **The Marshmallow Test:** A study on delayed gratification, conducted by Walter Mischel in the 1960s.

9. **The Invisible Gorilla:** A study on inattention blindness, conducted by Christopher Chabris and Daniel Simons in 1999.
10. **The Trolley Problem:** A thought experiment in ethics, first introduced by Philippa Foot in 1967.

1. Calibration of Analytical Equipment and apparatus

Calibration is the process of verifying the accuracy of an instrument or device by comparing its measurements with a known standard. This is important for ensuring that the results obtained from the instrument are accurate and reliable.

Here are some key points to consider when calibrating analytical equipment and apparatus:

- Calibration should be performed regularly to ensure the accuracy of the instrument over time.
- The frequency of calibration will depend on factors such as the type of instrument, its usage, and the environment in which it is used.
- Calibration should be performed using a reference standard that is traceable to a national or international standard.
- The calibration process should be documented, including the date of calibration, the reference standard used, and the results obtained.
- Any adjustments or corrections made to the instrument during calibration should also be documented.
- It is important to follow the manufacturer's instructions when calibrating an instrument to ensure that the process is performed correctly.
- In some cases, it may be necessary to send the instrument to a calibration laboratory for calibration by a qualified technician.

Overall, calibration is an essential step in ensuring the accuracy and reliability of analytical equipment and apparatus. By following the recommended procedures and documenting the process, you can have confidence in the results obtained from your instruments.

2. Determination of Hardness of water sample by EDTA method

- Hardness of water is determined by the presence of calcium and magnesium ions.
- The EDTA (ethylenediaminetetraacetic acid) method is a common method used to determine the hardness of water.
- In this method, a solution of EDTA is used to titrate a water sample.
- The EDTA binds with the calcium and magnesium ions present in the water, forming a complex.
- An indicator such as Eriochrome Black T is used to signal the end point of the titration.
- The amount of EDTA used in the titration is proportional to the hardness of the water sample.

- The hardness of the water sample can be calculated using the formula:
Hardness (mg/L) = (Volume of EDTA used * 1000) / Volume of water sample.
- This method is simple, accurate, and widely used in the determination of water hardness.

3. Determination of Alkalinity of water sample

Alkalinity is a measure of the ability of water to neutralize acids. It is an important parameter in water quality analysis, as it provides information about the buffering capacity of water.

Here are the steps to determine the alkalinity of a water sample:

1. Collect a water sample in a clean container.
2. Measure the pH of the water sample using a pH meter or pH test strips.
3. If the pH is above 7, the water sample is alkaline.
4. To determine the exact alkalinity of the water sample, titrate the sample with a standard acid solution, such as sulfuric acid or hydrochloric acid.
5. The volume of acid required to neutralize the water sample to a pH of 4.5 is used to calculate the alkalinity of the water sample.
6. The alkalinity is usually expressed in milligrams per liter (mg/L) of calcium carbonate (CaCO₃).

It is important to note that the alkalinity of water can vary depending on the source and the presence of minerals and other substances that can affect the buffering capacity of water. Therefore, it is recommended to regularly test the alkalinity of water to ensure its quality.

4. Determination of pH by titrimetric method

Titrimetric method is a common laboratory technique used to determine the pH of a solution. The process involves the following steps:

1. A solution of known concentration, called the titrant, is prepared. This is usually a strong acid or base.
2. The solution to be tested, called the analyte, is measured and placed in a container.
3. A few drops of an indicator are added to the analyte. The indicator is a substance that changes color depending on the pH of the solution.
4. The titrant is slowly added to the analyte while stirring. The indicator changes color when the pH of the solution reaches a certain value.
5. The volume of titrant added is recorded when the indicator changes color. This is called the endpoint.
6. The pH of the solution can be calculated using the volume of titrant added and the concentration of the titrant.

This method is accurate and reliable, but it requires careful preparation and execution. It is commonly used in laboratories and in industrial processes where precise pH measurements are required.

5. Determination of surface tension of given liquid

Surface tension is a property of liquids that arises due to the cohesive forces between the liquid molecules. It is defined as the force acting per unit length on the surface of the liquid. There are several methods to determine the surface tension of a given liquid, including:

1. **Capillary rise method:** This method involves measuring the height to which a liquid rises in a capillary tube due to the surface tension of the liquid. The surface tension is then calculated using the formula: $\text{surface tension} = (\text{weight of liquid raised} \times \text{radius of capillary tube}) / (\text{height of liquid rise} \times 2 \times \pi)$
2. **Maximum bubble pressure method:** This method involves measuring the maximum pressure required to blow a bubble from a submerged orifice in the liquid. The surface tension is then calculated using the formula: $\text{surface tension} = (4 \times \text{maximum pressure} \times \text{radius of orifice}) / 3$
3. **Drop weight method:** This method involves measuring the weight of a droplet of liquid that detaches from the end of a thin tube. The surface tension is then calculated using the formula: $\text{surface tension} = (\text{weight of droplet} \times \text{acceleration due to gravity}) / (2 \times \pi \times \text{radius of tube})$
4. **Wilhelmy plate method:** This method involves immersing a thin flat plate into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: $\text{surface tension} = \text{force} / (2 \times \text{perimeter of plate})$
5. **Du Nouy ring method:** This method involves immersing a thin ring into the liquid and measuring the force required to detach it from the liquid surface. The surface tension is then calculated using the formula: $\text{surface tension} = (\text{force} - \text{weight of ring in air} + \text{weight of ring in liquid}) / (2 \times \pi \times \text{radius of ring})$

These are some of the common methods used to determine the surface tension of a given liquid. Each method has its own advantages and limitations, and the choice of method depends on the specific requirements of the experiment.

6. Determination of Viscosity of a given liquid by viscometer

Viscosity is a measure of a fluid's resistance to flow. It is commonly referred to as the thickness of a fluid. A viscometer is an instrument used to measure

the viscosity of a fluid. Here are the steps to determine the viscosity of a given liquid by viscometer:

1. **Select the appropriate viscometer:** Different types of viscometers are available for different types of fluids and measurement ranges. Select the appropriate viscometer for the liquid to be measured.
2. **Calibrate the viscometer:** Before taking any measurements, the viscometer must be calibrated according to the manufacturer's instructions.
3. **Prepare the sample:** The liquid to be measured should be prepared according to the viscometer's instructions. This may include filtering, degassing, or temperature control.
4. **Load the sample:** The liquid sample is loaded into the viscometer according to the manufacturer's instructions.
5. **Take the measurement:** The viscometer is operated according to the manufacturer's instructions to take the viscosity measurement.
6. **Record the results:** The viscosity measurement is recorded, along with any relevant information such as temperature, sample preparation, and instrument calibration.
7. **Clean the viscometer:** After the measurement is complete, the viscometer should be cleaned according to the manufacturer's instructions to ensure accurate measurements in the future.

By following these steps, the viscosity of a given liquid can be accurately determined using a viscometer. It is important to carefully follow the manufacturer's instructions and to properly calibrate and maintain the viscometer to ensure accurate measurements.

Determination of the strength of Ferrous ammonium sulfate using external indicator

1. Ferrous ammonium sulfate, also known as Mohr's salt, is a double salt of iron sulfate and ammonium sulfate.
2. Its chemical formula is $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$.
3. It is commonly used as a primary standard in redox titrations due to its high purity and stability.
4. The strength of a solution of ferrous ammonium sulfate can be determined by titrating it against a standard solution of potassium permanganate (KMnO_4) using an external indicator.
5. The external indicator used in this titration is potassium ferricyanide ($\text{K}_3\text{Fe}(\text{CN})_6$).
6. The reaction between ferrous ammonium sulfate and potassium permanganate is a redox reaction, where potassium permanganate is reduced to

manganese sulfate (MnSO_4) and ferrous ammonium sulfate is oxidized to ferric ammonium sulfate ($(\text{NH}_4)_3\text{Fe}(\text{SO}_4)_3$).

7. The end point of the titration is indicated by the appearance of a blue color, due to the formation of Prussian blue ($\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$) when potassium ferricyanide reacts with the excess ferric ions.
8. The strength of the ferrous ammonium sulfate solution can be calculated using the balanced chemical equation and the volume of potassium permanganate solution used in the titration.

8. Determination of the strength of Potassium dichromate using internal indicator

1. Potassium dichromate is a primary standard substance that can be used to determine the strength of a solution.
2. The strength of potassium dichromate can be determined using an internal indicator.
3. An internal indicator is a substance that changes color at the end point of a titration.
4. In the case of potassium dichromate, the internal indicator used is usually diphenylamine sulfonate.
5. Diphenylamine sulfonate changes color from colorless to blue at the end point of the titration.
6. The end point of the titration is reached when the blue color persists for at least 30 seconds.
7. The strength of the potassium dichromate solution can then be calculated using the volume of the solution used and the concentration of the titrant.
8. This method is accurate and reliable for determining the strength of potassium dichromate solutions.

9. Determination of available chlorine in bleaching powder

Bleaching powder, also known as calcium hypochlorite, is a chemical compound that is commonly used as a disinfectant and bleaching agent. The available chlorine in bleaching powder refers to the amount of chlorine that can be released from the compound when it reacts with an acid.

To determine the available chlorine in bleaching powder, the following steps can be followed:

1. Weigh a sample of bleaching powder and dissolve it in water.
2. Add an excess of potassium iodide (KI) to the solution. The iodide ions will react with the available chlorine to produce iodine.
3. Titrate the iodine produced with a standard solution of sodium thiosulfate, using starch as an indicator. The starch will turn blue in the presence of iodine, indicating the end point of the titration.

4. Calculate the amount of available chlorine in the sample using the volume of sodium thiosulfate used in the titration and the concentration of the sodium thiosulfate solution.

This method is known as the iodometric titration method and is commonly used to determine the available chlorine in bleaching powder. It is important to note that the results obtained using this method may vary depending on factors such as the age and storage conditions of the bleaching powder sample.

Determination of Chloride Content in Water Sample

1. Chloride ions are one of the major inorganic anions in water and wastewater.
2. The chloride test measures the level of chloride ions in water.
3. Chloride ions can enter water sources through various means, including natural processes such as the weathering of rocks and human activities such as the use of road salt and the discharge of industrial effluents.
4. High levels of chloride ions in water can have negative impacts on aquatic life and can also affect the taste of drinking water.
5. The most common method for determining the chloride content in water is the argentometric method, also known as the Mohr method.
6. In this method, a measured volume of water sample is titrated with a standard solution of silver nitrate using potassium chromate as an indicator.
7. The end point of the titration is indicated by the formation of a red-brown precipitate of silver chromate.
8. The amount of silver nitrate used in the titration is used to calculate the concentration of chloride ions in the water sample.
9. Other methods for determining the chloride content in water include ion chromatography, ion-selective electrodes, and colorimetric methods.
10. Regular monitoring of the chloride content in water sources is important for maintaining the health of aquatic ecosystems and ensuring the quality of drinking water.

11. Preparation of Phenol formaldehyde (PF) resin

Phenol formaldehyde (PF) resin, also known as phenolic resin, is a synthetic polymer obtained by the reaction of phenol with formaldehyde. Here are the steps involved in the preparation of PF resin:

1. **Phenol and formaldehyde are mixed:** Phenol and formaldehyde are mixed in the presence of an acidic or basic catalyst. The ratio of formaldehyde to phenol is typically between 1.5:1 and 2:1.
2. **Formation of hydroxymethylphenols:** The reaction between phenol and formaldehyde produces hydroxymethylphenols, which can further re-

act with phenol to form dimers, trimers, and higher oligomers.

3. **Formation of resin:** The oligomers formed in the previous step can further react with each other to form a three-dimensional network, resulting in the formation of the resin.
4. **Curing:** The resin is then cured by heating it at a high temperature, typically between 120°C and 180°C, for several hours. This causes the resin to harden and become insoluble and infusible.

PF resin is widely used in the production of laminates, molding compounds, and wood adhesives due to its high mechanical strength, heat resistance, and chemical resistance. It is also used as a binder in the production of abrasive products and as an insulation material in the electrical and electronics industries.

12. Preparation of Urea formaldehyde (UF) resin

Urea formaldehyde (UF) resin is a type of thermosetting polymer that is commonly used in the production of particleboard, medium-density fiberboard (MDF), and other wood products. It is made by reacting urea and formaldehyde in the presence of a catalyst, typically an acid or a base.

Here are the steps involved in the preparation of UF resin:

1. Urea and formaldehyde are mixed in a reactor vessel in a molar ratio of approximately 1:1.5 to 2.5.
2. A catalyst, such as sulfuric acid or sodium hydroxide, is added to the mixture to initiate the reaction.
3. The mixture is heated to a temperature of around 90-100°C and stirred continuously.
4. The reaction is allowed to proceed for several hours until the desired degree of polymerization is achieved.
5. The resulting resin is cooled and neutralized with an appropriate base or acid to stop the reaction.
6. The resin is then filtered to remove any impurities and can be used as is or further processed to modify its properties.

It is important to note that the exact conditions and proportions of the reactants may vary depending on the desired properties of the final product. Additionally, other additives, such as fillers or pigments, may be added to the resin to improve its performance or appearance.

13. Preparation of Adipic acid / Paracetamol

Adipic acid and paracetamol are two different compounds with different chemical structures and uses. Here is some information on the preparation of each compound:

Adipic acid:

- Adipic acid is an organic compound with the formula $(\text{CH}_2)_4(\text{COOH})_2$.
- It is a white crystalline powder that is mainly used in the production of nylon.
- Adipic acid can be prepared by the oxidation of cyclohexanol or cyclohexanone with nitric acid.
- Another method for the preparation of adipic acid is the carbonylation of butadiene in the presence of a nickel catalyst.

Paracetamol:

- Paracetamol, also known as acetaminophen, is an analgesic and antipyretic drug used to relieve pain and reduce fever.
- It is prepared by the acetylation of para-aminophenol with acetic anhydride.
- The reaction takes place in the presence of an acid catalyst, such as sulfuric acid or hydrochloric acid.
- The resulting product is then purified by recrystallization from water.

14. Determination of Cell Conductance of a solution

1. Cell conductance is a measure of a solution's ability to conduct electricity.
2. It is determined by measuring the electrical conductivity of the solution using a conductivity meter.
3. The conductivity meter consists of a cell with two electrodes and a circuit to measure the current passing through the solution.
4. The cell constant is determined by measuring the conductance of a solution of known conductivity.
5. The conductance of the unknown solution is then measured and the cell constant is used to calculate the conductivity of the solution.
6. The conductivity of the solution is directly proportional to the concentration of ions in the solution.
7. The temperature of the solution must be taken into account when measuring conductivity, as it affects the mobility of the ions.
8. The units of conductivity are Siemens per meter (S/m) or microSiemens per centimeter ($\mu\text{S}/\text{cm}$).
9. The cell conductance of a solution can be used to determine the concentration of ions in the solution, the purity of water, and the salinity of seawater.

15. Determination of Rate constant of hydrolysis of esters

1. The rate constant of hydrolysis of esters can be determined by measuring the rate of reaction between an ester and water.
2. The reaction between an ester and water is known as hydrolysis and results in the formation of a carboxylic acid and an alcohol.
3. The rate of reaction can be measured by monitoring the change in concentration of one of the reactants or products over time.

4. The rate constant can then be calculated using the rate law for the reaction.
5. The rate law for the hydrolysis of esters is typically first order with respect to the ester and water.
6. This means that the rate of reaction is directly proportional to the concentration of the ester and water.
7. The rate constant can be determined by plotting the natural logarithm of the concentration of the ester against time and calculating the slope of the line.
8. The slope of the line is equal to the negative of the rate constant.
9. The units of the rate constant for a first-order reaction are typically expressed in reciprocal seconds (s^{-1}).
10. The rate constant provides information about the speed of the reaction and can be used to predict the rate of reaction under different conditions.

16. Element detection and identification of functional groups in organic compounds

Organic compounds are composed of carbon, hydrogen, and other elements such as nitrogen, oxygen, sulfur, and halogens. The presence of these elements can be detected through various tests.

1. **Detection of Carbon and Hydrogen:** Carbon and hydrogen are detected by heating the organic compound with copper oxide in a test tube. Carbon is oxidized to carbon dioxide and hydrogen is oxidized to water. The carbon dioxide turns lime water milky, while the water formed condenses on the cooler parts of the test tube.
2. **Detection of Nitrogen:** Nitrogen can be detected by the Lassaigne's test. In this test, the organic compound is fused with sodium metal. The nitrogen in the organic compound is converted to sodium cyanide, which can be detected by the formation of Prussian blue color with iron(III) chloride and iron(II) sulfate.
3. **Detection of Oxygen:** Oxygen is not directly detected in organic compounds. Instead, the presence of oxygen is inferred from the absence of nitrogen, sulfur, and halogens.
4. **Detection of Sulfur:** Sulfur can be detected by the sodium fusion test. In this test, the organic compound is fused with sodium metal. The sulfur in the organic compound is converted to sodium sulfide, which can be detected by the formation of a black precipitate with lead acetate.
5. **Detection of Halogens:** Halogens can be detected by the Beilstein test. In this test, the organic compound is heated with a copper wire. The halogens in the organic compound react with the copper to form copper halides, which impart a characteristic color to the flame.

Functional groups are specific groups of atoms within organic molecules that are

responsible for the characteristic chemical reactions of those molecules. Some common functional groups and their identification tests are:

1. **Alcohols:** Alcohols contain the hydroxyl (-OH) functional group. They can be identified by the Lucas test, in which the alcohol is treated with Lucas reagent (a mixture of zinc chloride and hydrochloric acid). Primary alcohols do not react, secondary alcohols form a cloudy solution, and tertiary alcohols form a separate layer.
2. **Aldehydes and Ketones:** Aldehydes and ketones contain the carbonyl ($>\text{C}=\text{O}$) functional group. They can be identified by the 2,4-dinitrophenylhydrazine (2,4-DNP) test, in which the aldehyde or ketone reacts with 2,4-DNP to form a yellow or orange precipitate.
3. **Carboxylic Acids:** Carboxylic acids contain the carboxyl (-COOH) functional group. They can be identified by their reaction with sodium bicarbonate to produce carbon dioxide gas.
4. **Amines:** Amines contain the amino (-NH₂) functional group. They can be identified by their reaction with nitrous acid to produce nitrogen gas.

These are some of the methods used for the detection of elements and identification of functional groups in organic compounds. It is important to note that these tests are not always conclusive and further analysis may be required for accurate identification.

NOTE: Instructor may choose any 10 experiments from above and may also change any two of the above.

- The instructor has the flexibility to choose any 10 experiments from the list provided.
- The instructor also has the option to change any two of the experiments from the list.
- This allows the instructor to tailor the experiments to the needs and interests of the students.
- It is important for students to carefully read the list of experiments and be prepared for any changes made by the instructor.
- Students should also communicate with the instructor to clarify any questions or concerns they may have about the experiments.

Course Outcomes:

1. Understanding of the subject matter: Students will gain a thorough understanding of the course material and be able to apply the concepts and theories to real-world situations.

2. Critical thinking: Students will develop the ability to analyze, evaluate, and synthesize information to make informed decisions and solve problems.
3. Communication skills: Students will improve their ability to communicate effectively, both orally and in writing, and to present their ideas clearly and persuasively.
4. Collaboration: Students will learn to work effectively in teams, to collaborate with others, and to build consensus.
5. Professionalism: Students will develop a strong sense of professionalism, including ethical behavior, responsibility, and accountability.
6. Lifelong learning: Students will develop the skills and habits necessary for lifelong learning, including the ability to adapt to new situations and to continue to grow and develop professionally.
7. Global perspective: Students will gain an understanding of global issues and the interconnectedness of the world, and will develop the ability to think and act globally.
8. Technological proficiency: Students will become proficient in the use of technology and will be able to use it effectively to enhance their learning and professional development.
9. Creativity and innovation: Students will develop the ability to think creatively and to generate new ideas and approaches, and will learn to apply innovative solutions to complex problems.
10. Leadership: Students will develop the skills and qualities necessary to become effective leaders, including the ability to inspire and motivate others, to set and achieve goals, and to make difficult decisions.

Upon completion of the course the student should be able to:

1. Demonstrate a comprehensive understanding of the subject matter.
2. Apply the knowledge and skills acquired in the course to real-world scenarios.
3. Analyze and evaluate information critically and effectively.
4. Communicate ideas and arguments clearly and effectively in written and oral forms.
5. Work collaboratively with others to achieve common goals.
6. Demonstrate ethical and professional behavior in academic and professional settings.
7. Engage in lifelong learning and professional development.

Course Outcomes Bloom's

Course outcomes are statements that describe the knowledge, skills, and abilities that students should have acquired by the end of a course. These outcomes

are typically aligned with the course objectives and are used to assess the effectiveness of the course in achieving its goals.

Bloom's Taxonomy is a framework for categorizing educational goals and objectives into different levels of complexity and specificity. It is often used to develop course outcomes and to design assessments that measure student achievement of those outcomes.

The taxonomy consists of six levels, arranged in a hierarchy from lower-order thinking skills to higher-order thinking skills:

1. **Remembering:** The ability to recall or retrieve previously learned information.
2. **Understanding:** The ability to comprehend the meaning of material.
3. **Applying:** The ability to use learned material in new and concrete situations.
4. **Analyzing:** The ability to break down material into its component parts so that its organizational structure may be understood.
5. **Evaluating:** The ability to make judgments about the value of ideas or materials.
6. **Creating:** The ability to put parts together to form a new whole.

By using Bloom's Taxonomy to develop course outcomes, instructors can ensure that their courses are designed to promote higher-order thinking skills and to provide students with opportunities to demonstrate their mastery of the course material.

Level

- A level is a tool used to determine if a surface is horizontal (level) or vertical (plumb).
- Levels can be used in construction, carpentry, surveying, and many other applications.
- There are several types of levels, including spirit levels, laser levels, and water levels.
- Spirit levels use a liquid-filled vial with an air bubble to indicate levelness. The bubble will be centered between two lines when the surface is level.
- Laser levels project a laser beam to create a straight line, which can be used to determine levelness.
- Water levels use the principle that water will always find its own level. A clear tube filled with water can be used to determine if two points are at the same level.
- It is important to use a level when installing floors, hanging shelves, or performing other tasks that require a level surface. A surface that is not level can cause problems such as uneven floors or doors that do not close properly.

CO-1 Get an understanding of the use of different analytical instruments. K3

Analytical instruments are devices used to measure, analyze, and quantify the physical and chemical properties of a sample. These instruments are commonly used in scientific research, quality control, and medical diagnosis. Some of the most commonly used analytical instruments include:

1. **Spectrophotometers:** These instruments measure the absorption or transmission of light by a sample, and are used to determine the concentration of a substance in a solution.
2. **Chromatographs:** These instruments separate and analyze the components of a mixture, and are commonly used in chemical and pharmaceutical analysis.
3. **Mass spectrometers:** These instruments measure the mass-to-charge ratio of ions, and are used to identify and quantify the components of a sample.
4. **Microscopes:** These instruments magnify small objects, and are used to observe and analyze the structure of cells, tissues, and other materials.
5. **Thermal analyzers:** These instruments measure the physical and chemical changes that occur in a sample as it is heated or cooled, and are used to study the thermal properties of materials.

Understanding the use of these and other analytical instruments is essential for conducting accurate and reliable scientific research and analysis. By selecting the appropriate instrument for a given task, scientists can obtain precise and accurate data to support their conclusions.

CO-2 Measure the molecular / system properties such as surface tension

Surface tension is a property of liquids that arises due to the cohesive forces between the molecules at the surface of the liquid. These forces cause the liquid to behave as if it is covered by an elastic film, which resists deformation and tends to minimize the surface area of the liquid.

Some key points to remember when measuring surface tension are:

1. Surface tension is typically measured in units of force per unit length, such as millinewtons per meter (mN/m).
2. There are several methods for measuring surface tension, including the drop weight method, the maximum bubble pressure method, and the Wilhelmy plate method.
3. The choice of method will depend on the specific properties of the liquid being measured, as well as the desired accuracy and precision of the measurement.
4. It is important to ensure that the liquid is clean and free of contaminants, as these can affect the surface tension measurement.

5. Temperature can also affect surface tension, so it is important to control and record the temperature during the measurement.

Viscosity, Conductance of Solution, Chloride and Iron Content in the Water

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- The viscosity of formation water is a function of pressure, temperature, and dissolved solids.
- In general, brine viscosity increases with increasing pressure, increasing salinity, and decreasing temperature.
- Dissolved gas in the formation water at reservoir conditions generally results in a negligible effect on water viscosity .

Conductance of Solution

- Conductivity is a measure of a solution's ability to conduct an electric current.
- The more chloride ions present in the water, the higher the conductivity.
- In general, the conductivity of waterways in the United States varies from 50 to 1500 $\mu\text{mhos}/\text{cm}$, and inland freshwater lake studies reveal a conductivity of 150 to 500 $\mu\text{mhos}/\text{cm}$.
- High-quality deionized water has a conductivity of about 0.05 S/cm at 25 °C, typical drinking water is in the range of 200–800 S/cm, while seawater is about 50 mS/cm (or 0.05 S/cm) .

Chloride Content in the Water

- Chloride ions are approximately 1.9 percent of the mass of seawater .

Iron Content in the Water

K3

K3 is a term that can refer to several different things. Without more context, it is difficult to determine which specific topic you are referring to. Some possible interpretations of K3 include:

1. K3 is a Belgian-Dutch girl group that was formed in 1998. The group is primarily known for their music and television series, which are aimed at children.
2. K3 is a visa type issued to spouses of U.S. citizens who are waiting for their immigrant visa petitions to be approved.

3. K3 is a mountain in the Karakoram range, located in the Gilgit-Baltistan region of Pakistan. It is also known as Broad Peak.
4. K3 is a type of tax form used in the United States to report income from estates and trusts.

CO-3 Measure the hardness and alkalinity of the water. K3

Hardness and alkalinity are two important parameters of water quality. Hardness refers to the concentration of calcium and magnesium ions in water, while alkalinity refers to the ability of water to neutralize acids.

1. **Measuring Hardness:** There are several methods to measure the hardness of water, including titration, colorimetry, and test strips. The most common method is titration, where a solution of ethylenediaminetetraacetic acid (EDTA) is used to titrate the water sample. The endpoint of the titration is indicated by a color change, and the amount of EDTA used is proportional to the hardness of the water.
2. **Measuring Alkalinity:** Alkalinity can be measured using titration as well. In this case, a solution of a strong acid, such as hydrochloric acid (HCl), is used to titrate the water sample. The endpoint of the titration is indicated by a pH meter or a color change, and the amount of acid used is proportional to the alkalinity of the water.

It is important to measure the hardness and alkalinity of water to ensure that it is suitable for its intended use. For example, water with high hardness can cause scaling in pipes and appliances, while water with low alkalinity can be corrosive. By measuring these parameters, appropriate treatment can be applied to improve the quality of the water.

CO-4 Know the fundamental concepts of the preparation of phenol

Phenol is an organic compound with the chemical formula C_6H_5OH . It is a white crystalline solid that is volatile and has a characteristic odor. Phenol is commonly used as a disinfectant and antiseptic, and is also used in the production of various chemicals, including plastics, resins, and synthetic fibers.

There are several methods for the preparation of phenol, including:

1. **From Cumene:** Cumene is oxidized in the presence of air to produce cumene hydroperoxide, which is then cleaved in the presence of an acid catalyst to produce phenol and acetone.
2. **From Chlorobenzene:** Chlorobenzene is heated with sodium hydroxide at high temperatures and pressures to produce phenol and sodium chloride.
3. **From Benzene:** Benzene is first converted to nitrobenzene by nitration, and then reduced to aniline. Aniline is then diazotized and treated with water to produce phenol.

4. **From Benzene Sulphonic Acid:** Benzene is first sulfonated to produce benzene sulphononic acid, which is then fused with sodium hydroxide to produce phenol and sodium sulphate.

These are some of the fundamental concepts of the preparation of phenol. It is important to understand these methods and the underlying chemical reactions in order to effectively produce phenol for various industrial and medical applications.

Formaldehyde, Urea Formaldehyde Resin, Adipic Acid, and Paracetamol

Formaldehyde

- Formaldehyde is a colorless, flammable, strong-smelling chemical that is used in building materials and to produce many household products.
- It is used in pressed-wood products, such as particleboard, plywood, and fiberboard; glues and adhesives; permanent-press fabrics; paper product coatings; and certain insulation materials.

Urea Formaldehyde Resin

- Urea-formaldehyde (UF), also known as urea-methanal, is a nontransparent thermosetting resin or polymer.
- It is produced from urea and formaldehyde.
- These resins are used in adhesives, plywood, particle board, medium-density fibreboard (MDF), and molded objects.
- Using formaldehyde and urea as raw materials, a stable urea-formaldehyde resin (UF) is synthesized by the “alkali-acid-alkali” method .
- Unlike most thermosetting resins, UF often shows the appearance of crystal domains .

Adipic Acid

- Adipic acid dihydrazide can be used as a modifier to reduce the free formaldehyde in urea-formaldehyde (UF) resin.
- The influence of the addition amount and addition stage of modifiers on the physicochemical property and free formaldehyde content of UF resin can be tested.

Paracetamol

K3

- K3 is a term that can refer to several different things, depending on the context in which it is used.

- It could refer to a mountain in the Karakoram range, also known as Broad Peak.
- It could also refer to a visa category for spouses of U.S. citizens.
- In mathematics, K3 surfaces are a type of algebraic surface.
- In music, K3 is a Belgian-Dutch girl group.
- In transportation, K3 may refer to a class of passenger carriages used by the Indian Railways.

CO-5 Estimate the rate constant of reaction. K3

The rate constant of a reaction, denoted by the symbol k , is a measure of the speed of a chemical reaction. It is defined as the proportionality constant between the rate of the reaction and the concentrations of the reactants raised to their respective powers. The value of k is dependent on the temperature, pressure, and the presence of a catalyst.

To estimate the rate constant of a reaction, the following steps can be followed:

1. Conduct a series of experiments to measure the rate of the reaction at different concentrations of the reactants.
2. Plot the data on a graph with the rate of the reaction on the y-axis and the concentration of the reactants on the x-axis.
3. Determine the order of the reaction with respect to each reactant by analyzing the shape of the graph.
4. Use the integrated rate law for the determined reaction order to calculate the value of the rate constant, k .

It is important to note that the value of k is specific to the reaction conditions and may change if the temperature, pressure, or the presence of a catalyst is altered. Therefore, it is necessary to specify the reaction conditions when reporting the value of k .

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- I'm sorry, but I don't have enough information to write about the topic "Page 33 of 40".
- Could you please provide more context or details about the topic you want me to write about?
- Is "Page 33 of 40" a title or a reference to a specific page in a book or document?
- Providing more information will help me generate a more accurate and informative response.

ENGINEERING GRAPHICS LAB KCS

Engineering Graphics is a subject that helps students in many ways for making images, describing the shape, size, finish, color, and construction of any object . It is the language of engineers and is meant for communicating their ideas, thoughts, and designs to others .

Some of the topics that are covered in an Engineering Graphics Lab course include: - Engineering drawing as a means of communication - Principles of engineering graphics - Freehand sketching - Introduction to AutoCAD and AutoCAD commands - Engineering drawing with AutoCAD - Orthographic projection - Lines and dimensioning - Reading blueprints - Normal, inclined, and cylindrical surfaces - Sectional views

Course Objectives:

- To provide students with a comprehensive understanding of the subject matter.
- To equip students with the necessary skills and knowledge to apply the concepts learned in real-world situations.
- To encourage critical thinking and problem-solving skills.
- To foster an appreciation for the subject and its relevance in today's world.
- To prepare students for further study or career advancement in the field.
- To promote active engagement and participation in the learning process.
- To assess student progress and provide feedback to facilitate improvement.
- To create a supportive and inclusive learning environment.

Fundamental Concepts of Analytical Instruments

Analytical instruments are devices used to measure, analyze, and determine the physical and chemical properties of substances. These instruments are essential tools in scientific research, medical diagnosis, and industrial processes.

Here are some fundamental concepts of analytical instruments:

1. **Accuracy:** This refers to how close the measurement is to the true value. An accurate instrument will provide measurements that are close to the true value of the substance being measured.
2. **Precision:** This refers to the consistency of the measurements. A precise instrument will provide consistent measurements, even if they are not necessarily close to the true value.
3. **Sensitivity:** This refers to the ability of the instrument to detect small changes in the substance being measured. A sensitive instrument will be able to detect even small changes in the physical or chemical properties of the substance.

4. **Selectivity:** This refers to the ability of the instrument to distinguish between different substances. A selective instrument will be able to accurately measure the properties of one substance, even in the presence of other substances.
5. **Calibration:** This is the process of adjusting the instrument to ensure that it provides accurate and precise measurements. Calibration is essential to ensure that the instrument is functioning correctly and providing reliable results.
6. **Signal-to-Noise Ratio:** This refers to the ratio of the signal (the measurement) to the noise (unwanted variations in the measurement). A high signal-to-noise ratio indicates that the instrument is providing a clear and reliable measurement.

These are some of the fundamental concepts of analytical instruments. Understanding these concepts is essential for students to effectively use and interpret the results of these instruments.

Analysis of Chloride Content, Hardness, and Alkalinity

1. **Chloride Content:** Chloride content refers to the amount of chloride ions present in a sample of water. Chloride ions are negatively charged particles that can be found in many types of salts, including sodium chloride (common table salt). The presence of chloride ions in water can indicate contamination from sources such as seawater intrusion, industrial waste, or road salt runoff.
2. **Hardness:** Water hardness refers to the concentration of calcium and magnesium ions in water. These ions can cause problems in industrial and domestic settings by forming deposits on pipes and fixtures, interfering with the action of soap and detergents, and reducing the efficiency of water heaters. Hardness is typically measured in grains per gallon (gpg) or milligrams per liter (mg/L) as calcium carbonate.
3. **Alkalinity:** Alkalinity is a measure of the ability of water to neutralize acids. It is primarily determined by the presence of carbonate, bicarbonate, and hydroxide ions in water. Alkalinity is important in maintaining the pH balance of water and preventing corrosion in pipes and fixtures. It is typically measured in milligrams per liter (mg/L) as calcium carbonate.

These three parameters are important in the analysis of water quality and can provide valuable information about the sources of contamination and the suitability of water for various uses. It is important for students to understand the methods used to measure these parameters and the significance of the results obtained.

Water

- Water is a substance composed of the chemical elements hydrogen and oxygen and existing in gaseous, liquid, and solid states.
- It is one of the most plentiful and essential of compounds.
- A tasteless and odorless liquid at room temperature, it has the important ability to dissolve many other substances.
- Water is a tiny molecule. It consists of three atoms: two of hydrogen and one of oxygen.
- Water molecules cling to each other because of a force called hydrogen bonding.
- Water is a shape-shifter. It exists in three states on Earth: liquid, gas, and solid.
- Water is the liquid that makes life on Earth possible.
- As water cycles from the air to the land to the sea and back again, water shapes our planet and nearly every aspect of our lives.
- All living things, from tiny cyanobacteria to giant blue whales, need water to survive.
- Water is an inorganic compound with the chemical formula H_2O .
- It is a transparent, tasteless, odorless, and nearly colorless chemical substance.
- It is the main constituent of Earth's hydrosphere and the fluids of all known living organisms (in which it acts as a solvent).

3. pH, Surface Tension, and Viscosity

pH

- pH is a measure of the acidity or basicity of a solution.
- It is defined as the negative logarithm of the hydrogen ion concentration in a solution.
- The pH scale ranges from 0 to 14, with 7 being neutral.
- A pH value less than 7 indicates an acidic solution, while a pH value greater than 7 indicates a basic solution.

Surface Tension

- Surface tension is the property of a liquid that allows it to resist an external force.
- It is caused by the cohesive forces between the molecules of the liquid.
- Surface tension is measured in units of force per unit length, typically in millinewtons per meter (mN/m).
- The surface tension of a liquid can be affected by various factors, including temperature, the presence of surfactants, and the nature of the liquid itself.

Viscosity

- Viscosity is a measure of a fluid's resistance to flow.
- It is defined as the ratio of the shear stress to the shear rate in a fluid.
- Viscosity is typically measured in units of pascal-seconds ($\text{Pa} \cdot \text{s}$).
- The viscosity of a fluid can be affected by various factors, including temperature, pressure, and the presence of impurities.

Liquid

- A liquid is one of the four fundamental states of matter, along with solid, gas, and plasma.
- It is characterized by its ability to flow and take the shape of its container.
- Liquids have a definite volume, but not a definite shape.
- The molecules in a liquid are closely packed, but they have enough energy to move around and slide past each other.
- The attractive forces between the molecules in a liquid are weaker than in a solid, but stronger than in a gas.
- The temperature at which a substance changes from a solid to a liquid is called its melting point, and the temperature at which it changes from a liquid to a gas is called its boiling point.
- Liquids can be classified as Newtonian or non-Newtonian, depending on whether their viscosity changes with the applied shear stress.
- Some common examples of liquids include water, milk, oil, and gasoline.
- Liquids play an important role in many natural and industrial processes, such as weather patterns, biological functions, and chemical reactions.

4. Preparation of Different Resins

Resins are solid or highly viscous substances that are typically convertible into polymers. They are used in a variety of applications, including adhesives, coatings, and plastics. Here are some points to consider when preparing different resins:

1. **Type of resin:** There are several types of resins, including natural resins (such as rosin and shellac) and synthetic resins (such as phenolic and epoxy resins). The type of resin used will depend on the desired properties and application.
2. **Raw materials:** The raw materials used to prepare the resin will vary depending on the type of resin being prepared. For example, phenolic resins are typically prepared from phenol and formaldehyde, while epoxy resins are prepared from epichlorohydrin and bisphenol A.
3. **Reaction conditions:** The reaction conditions, such as temperature, pressure, and catalyst, will also vary depending on the type of resin be-

ing prepared. For example, the preparation of phenolic resins typically involves heating the reactants in the presence of an acid catalyst.

4. **Purification:** After the resin has been prepared, it may need to be purified to remove any impurities or unreacted raw materials. This can be done through methods such as filtration, distillation, or extraction.
5. **Formulation:** The final step in the preparation of resins is formulation, where the resin is mixed with other ingredients, such as fillers, pigments, and additives, to achieve the desired properties and performance in its intended application.

In summary, the preparation of different resins involves selecting the appropriate type of resin, using the correct raw materials and reaction conditions, purifying the resin, and formulating it with other ingredients to achieve the desired properties and performance. It is important to carefully control each step of the process to ensure that the final resin meets the desired specifications.

Synthesis of Adipic Acid

Adipic acid is an important organic compound that is used in the production of nylon, polyurethane, and other industrial products. Here are some key points to understand about the synthesis of adipic acid:

1. Adipic acid is a dicarboxylic acid with the chemical formula $C_6H_{10}O_4$.
2. It can be synthesized from a variety of starting materials, including cyclohexanol, cyclohexanone, and benzene.
3. One common method for synthesizing adipic acid is through the oxidation of cyclohexanol or cyclohexanone using nitric acid.
4. Another method involves the oxidation of benzene using a mixture of nitric and sulfuric acids.
5. The resulting adipic acid can be purified through recrystallization or other separation techniques.

These are some of the key points to understand about the synthesis of adipic acid. It is an important industrial chemical with a variety of uses, and its synthesis can be achieved through several different methods.

Acid and Paracetamol

- Mefenamic Acid + Paracetamol is a combination of two medicines: Mefenamic Acid and Paracetamol. It works by blocking the release of certain chemical messengers that cause fever, pain and inflammation (redness and swelling).
- Mefenamic Acid+Paracetamol is used for pain relief and fever. It relieves pain and inflammation in conditions like rheumatoid arthritis, osteoarthritis and menstrual pain. It is also used to treat mild to moderate pain.

- Paracetamol may be made by acetylation of para-aminophenol (obtained by reduction of para-nitrophenol) with acetic acid or acetic anhydride.
- Paracetamol is used as an analgesic and antipyretic drug. It is the preferred alternative analgesic-antipyretic to aspirin (acetylsalicylic acid), particularly in patients with coagulation disorders, individuals with a history of peptic ulcer or who cannot tolerate aspirin, as well as in children.
- Paracetamol (Panadol, Calpol, Alvedon) is an analgesic and antipyretic drug that is used to temporarily relieve mild-to-moderate pain and fever. It is commonly included as an ingredient in cold and flu medications and is also used on its own. Paracetamol is exactly the same drug as acetaminophen (Tylenol).